

Volcanic Forcing and the Pre-Industrial Malthusian Trap: Demographic Response, Boserupian Intensification, and the Long Shadow of Seasonality

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Abstract

We assemble a new monthly paleo-economic panel covering 196 countries from 1421 CE—ModE-RA $\times$ HYDE $\times$ eVolv2k $\times$ Brecke $\times$ Allen—and use it to identify three findings about the pre-industrial Malthusian trap. First, continuous volcanic forcing 1500–1900 produces a pathway-heterogeneous demographic response, with crop-dominant late and pastoral/mixed late countries losing population at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, Bonferroni-significant across multiple-testing and robustness slices. Second, the same crop-dominant late countries expanded cropland share after cold shocks in the modern instrumental era ($+7.8 \cdot 10^{-4}$ per Tg in 1800–1900, $p < 0.01$), the first cross-country pre-industrial test of the Boserupian intensification hypothesis using exogenous climate forcing. Third, pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, surviving latitude and deep-determinant controls. Seasonality, properly measured, casts a long shadow.

JEL Codes: N50, O13, O44, Q54, J11.

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1 Introduction

Most of human history was Malthusian. Productivity gains were absorbed by population growth; living standards drifted within a narrow band. Then, beginning in eighteenth-century northwestern Europe, the trap broke. Income per capita rose without compensating fertility, and the modern era of sustained growth began. Why did some economies escape while others stayed inside? Unified growth theory has converged on a short list of candidate answers—human capital accumulation (Galor and Weil, 2000), late marriage and urban disease in northwestern Europe (Voigtländer and Voth, 2013), gradual selection on individual industriousness (Clark, 2007), New-World-crop relaxation of the Malthusian ceiling (Nunn and Qian, 2011), and the institutional inheritances that translated productivity into welfare rather than headcount (Ashraf and Galor, 2013)—each of which has a climate sub-text but none of which has been tested against the actual climate that pre-industrial societies experienced.

The data needed to do that empirical test have not previously been assembled. Pre-industrial population and land use have been mapped at the country or continent level by historical demographers for decades but only recently at finer resolution; HYDE 3.5 (Klein Goldewijk et al., 2017) now

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provides 5-arcminute estimates of population, cropland, grazing, irrigation, and urban population for 197 countries from 10,000 BCE to 2025 CE. Climate has been harder. The PAGES 2k continental reconstructions (PAGES 2k Consortium, 2019) organise the proxy record at decadal-to-continental resolution—adequate for global narratives, inadequate for what a Strasbourg wheat harvest looked like in 1437. The breakthrough we exploit is ModeE-RA (Franke et al., 2023), a monthly global paleo-reanalysis covering 1421–2008 CE that assimilates documentary records, tree rings, ice cores, corals, and historical instruments into a coupled numerical model. Combined with the Sigl-Toohey eVolv2k ice-core record of volcanic stratospheric sulfur injection (Sigl and Toohey, 2024; Toohey and Sigl, 2017), the Brecke conflict catalogue (Brecke, 1999), and the Allen-Nuffield archive of city-level wages and prices (Allen, 2001), ModeE-RA delivers the regressors that pre-industrial Malthusian econometrics has been missing.

We build a new monthly paleo-economic panel from these sources covering 196 countries and 3,187 sub-national units, validate its alignment with the modern instrumental record, and use it to identify three substantive findings. The empirical apparatus is, we believe, the more durable contribution: it makes within-country Malthusian regressions with real annual climate forcing addressable for the full 1421–1950 sample, sidesteps the country-level confounders that have shadowed cross-country growth empirics (Durlauf et al., 2005) when the within-country specification is identified off sub-national variation, and supports extensions to the 1257 Samalas eruption using city-level wheat prices and a region-aware exposure measure.

The substantive findings are three. First, continuous volcanic forcing from the eVolv2k ice-core record 1500–1900 produces a pathway-heterogeneous demographic response that survives every robustness exercise we run on it. Crop-dominant late and pastoral/mixed late countries show population-growth coefficients of -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, both significant under Bonferroni correction across twelve pathway-by-equation tests, both robust to re-clustering pathways on climate primitives that are orthogonal to HYDE outcomes, and both surviving restriction to the post-1700 decadal portion of HYDE where the demographic back-projection is less dependent on cropland imputation. This is the cleanest pre-industrial volcanic-cooling demographic-response coefficient that has been estimated at country resolution. Second, in the same panel restricted to the modern instrumental era, crop-dominant late countries show a sharply *positive* cropland-share response to volcanic forcing ($+7.8 \cdot 10^{-4}$ per Tg in 1800–1900, $p < 0.01$): after cold shocks, these economies expanded cropland. This is, to our knowledge, the first cross-country pre-industrial test of the Boserupian intensification hypothesis (Boserup, 1965) using exogenous climate forcing, and the sign and significance are robust across temporal sub-samples. Third, pre-industrial inter-annual climate volatility 1421–1750 emerges as the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, surviving both latitude and an extended deep-determinant battery; modern climate volatility is irrelevant for the same outcome ($p = 0.16$), a placebo that is hard to reproduce under any alternative interpretation.

Three corroborating exercises target the substantive concerns the headline regressions leave open. The Allen real-wage panel for six European countries 1421–1850 returns the canonical Malthusian price-on-population coefficient at city resolution, identifying the welfare margin that population alone cannot speak to. An annual European mortality and fertility panel 1751–1900 decomposes the Malthusian response into preventive (fertility) and positive (mortality) checks and finds it dominated by the positive check on infant and child mortality, the historically documented summer-mortality pattern of pre-modern Europe; the same coefficient collapses monotonically to zero by 1950–2022 as twentieth-century sanitation, refrigeration, and antibiotics develop, consistent with a sanitation-mediated mechanism. And a region-aware extension of the volcanic test to the 1257 Samalas eruption on city-level wheat prices recovers a supply-side spike at lag two years that survives both conflict-and-pandemic controls and hemispheric exposure weighting.

A joint three-equation panel VAR of population, cropland-share, and urban-share growth on the same exogenous volcanic forcing produces nominally heterogeneous adjustment margins across pathways, but the asymmetric three-margin pattern does not survive multiple-testing correction or sub-sample stress tests. We report it because the demographic-margin coefficients estimated within the joint specification are the same coefficients that pass the robustness exercises above; we treat the joint specification as the natural multi-equation generalisation of the Malthusian regression rather than as a multi-margin decomposition. Appendix J reports the three robustness exercises (climate-only pathway re-clustering, wild-cluster bootstrap and permutation inference, post-1700 sub-sample) that motivate this framing in detail.

The intellectual frame is not original to us. Galor (2011) provides unified growth theory; Mastranga (2024) provides the climate-drives-agriculture mechanism we extend; Ashraf and Galor (2011) demonstrates that the strength of the Malthusian feedback can be estimated directly from pre-industrial cross-country panels and that the variation it identifies is rich enough to discriminate between models, but in their setting climate forcing is a residual term inside the country fixed effect, not a measurable regressor. Dell et al. (2012); Burke et al. (2015) establish that temperature does have first-order effects on economic activity in the modern instrumental record, but the modern panel is short and post-industrial; the relevant question for unified growth theory is what climate forcing did during the centuries when Malthusian feedback was active. Our contribution is to bring real annual climate forcing to the pre-industrial period for the first time at country-monthly resolution, identify three robust empirical regularities about how that forcing operated on demographic and land-use outcomes, and surface a Boserupian intensification response that the existing literature has documented theoretically but not empirically.

The paper proceeds as follows. Section 2 introduces the new paleo-economic panel and the five-pathway agricultural typology. Section 3 works out the within-country identification strategy and runs the storage-demand sub-national regression. Section 4 runs the volcanic-forcing regression that delivers the first two substantive findings and the corroborating exercises that address welfare and demographic-mechanism concerns. Section 5 runs the long-shadow regression. Section 6 consolidates robustness; Section 7 concludes. The appendix collects the theoretical framework, the calibrated quantitative model, the pre-XV-century volcanic extensions, the war-and-pandemic robustness, a CamPop family-reconstitution annual series 1541–1850, and the three robustness exercises that motivate the framing of Section 4.

2 A new monthly paleo-economic panel

The paper rests on a paleo-economic dataset that we assemble for the first time and that does most of the empirical work. We describe its construction, validate its alignment with the modern instrumental record where they overlap, and then use it to characterise the climate-pathway typology that organises the rest of the paper. The dataset itself is, we believe, the more durable contribution: it can support econometric questions about pre-industrial development that the existing climate-history literature has been unable to address.

Temporal coverage of the data sources assembled into the paleo-economic panel

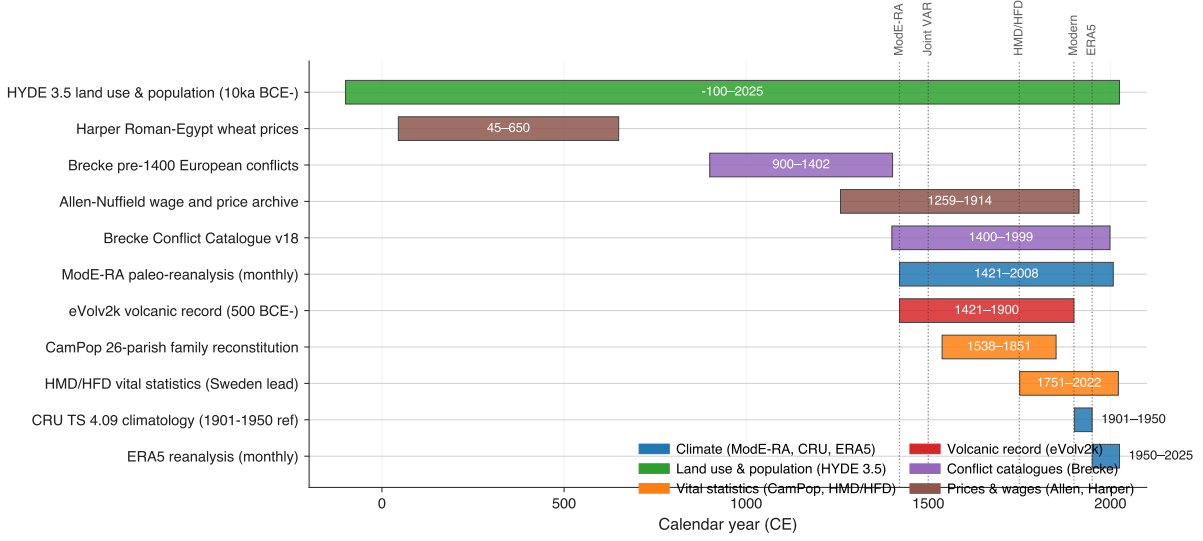


Figure 1: Temporal coverage of the data sources merged into the paleo-economic panel. Each bar marks one source. The vertical dotted lines mark the analysis windows (1421 ModE-RA start, 1500 joint VAR start, 1750 HMD/HFD-era start, 1900 modern, 1950 ERA5).

2.1 Construction

The dataset has three substantive layers and three event layers, all merged onto a common country-year (or city-year, or sub-unit-year) grid. The first substantive layer is HYDE 3.5 (Klein Goldewijk et al., 2017), which provides gridded estimates of population, cropland, grazing land, irrigated land, rice cultivation, and urban population at 5-arcminute resolution from 10,000 BCE to 2025 CE under three uncertainty scenarios. We aggregate to country totals using the `iso_cr` country raster and to 3,187 sub-national units using the `sub_iso_cr` raster. The HYDE temporal grid is irregular: 100-year intervals before 1700, decadal 1700–1950, and annual thereafter. The pre-industrial decadal sample is what powers most of the regressions; we use the century-spaced pre-1500 sample only in the appendix extension.

The second layer is ModE-RA (Franke et al., 2023), a monthly global paleo-reanalysis covering 1421–2008 CE that assimilates documentary records, tree rings, ice cores, corals, and historical instruments into a coupled numerical-model background, returning monthly temperature and precipitation anomalies relative to the 1901–2000 mean on a 192×96 Gaussian grid. We aggregate each ModE-RA cell onto HYDE country and sub-national units via cosine-latitude-weighted area sums on the 5-arcminute country raster; the CRU TS 4.09 (Harris et al., 2020) 1901–1950 country-month climatology, computed analogously, is added back to the anomalies to obtain absolute monthly temperatures and precipitation throughout 1421–2008. We rebuild ERA5 (Hersbach et al., 2020) at the country level by area-weighting the raw 0.25° ERA5 monthly fields onto HYDE country boundaries; per-country mean bias is removed by a constant offset estimated in the 1950–2008 overlap with ModE-RA.

Three derived climate measures play headline roles in what follows. The intra-annual temperature range $\sigma_s^T(i, t) = \max_m T_{i,t,m} - \min_m T_{i,t,m}$ captures within-year variation. The productive-months index $\text{Prod}_{i,t}$ counts calendar months in which $5^\circ\text{C} \leq T_{i,t,m} \leq 30^\circ\text{C}$ and $P_{i,t,m} \geq 30$ mm—months simultaneously warm enough and wet enough for rain-fed cropping; the complement $12 -$

$\text{Prod}_{i,t}$ is the non-productive-months count and captures storage demand. The inter-annual temperature volatility $\sigma_v^T(i, t)$ is the country-specific standard deviation of annual mean temperature in a 30-year centred window. In within-interval panel regressions we use within-interval volatility rather than the rolling window. We report a published-criterion alternative based on FAO-style wheat suitability in Section 6.

The three event layers attach to specific (country, year) or (city, year) cells of the panel. The Sigl-Toohey eVolv2k v4 ice-core record (Sigl and Toohey, 2024; Toohey and Sigl, 2017) provides annual estimates of global stratospheric volcanic sulfur injection from 500 BCE to 1900 CE, with eruption latitudes for each event; we link this to the climate panel as both a discrete event-study indicator and a continuous distributed-lag regressor. The Brecke Conflict Catalogue (Brecke, 1999) adds 3,708 violent conflicts 1400–1999 with start- and end-years and free-text actor descriptions, plus 1,147 European conflicts 900–1402 in a supplementary extension. We string-match the actor descriptions to nine medieval regions to produce binary war indicators per (city, year), and hand-code 30 sieges of specific panel cities and 49 plague episodes from standard catalogues. The Allen-Nuffield wage-and-price archive (Allen, 2001) contributes silver-content-normalised wheat prices for 17 European cities 1259–1914 and real-wage series for 8 cities 1264–1850; we parse the silver-grams-per-hectoliter wheat-price column directly from each city’s xls file and augment with Tuscan prices from Malanima (2002) and the Roman-Egypt wheat-price series of Harper (2016).

2.2 Validation

The country-level monthly climate panel matches the modern instrumental record where they overlap. In the 1950–2008 Mode-RA/ERA5 overlap (Figure 2), the median country-month mean bias is 0.61°C , the median country-monthly T correlation is 0.971 (77.6% of countries exceed $r = 0.90$), and the median country-annual anomaly correlation is 0.938 after detrending. The agreement is strong enough that we treat Mode-RA + CRU climatology as a substantive measurement of country-monthly climate rather than as a stand-in. Mode-RA’s ensemble 1σ uncertainty in country-month temperature, propagated through the area-weighted aggregation, is on the order of 0.9°C in 1421–1500 and falls steadily to 0.37°C by 1850–1949; Mode-RA precipitation uncertainty traces the same secular decline from 31.6 to 15.4 mm/month. Past 1950, we splice in our country-level ERA5 rebuild, which the bias-correction step described above brings onto the same scale as the paleo-reanalysis. Figure 10 reports the merged-panel uncertainty across 1421–2025, including the ERA5–Mode-RA monthly RMSE in the 1950–2008 overlap as a cross-validation diagnostic. Section 6 discusses the implications for the headline regressions.

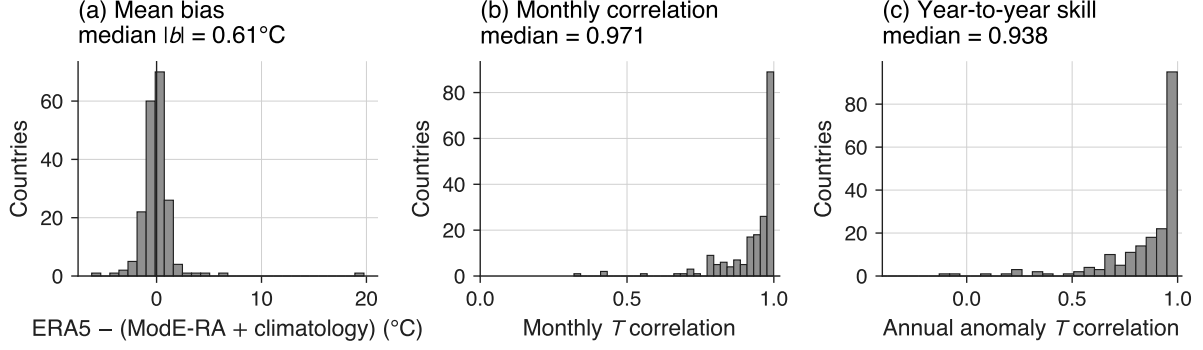


Figure 2: ModE-RA versus country-level ERA5, 1950–2008. (a) Per-country mean bias (ERA5 minus ModE-RA + CRU climatology). (b) Country-month temperature correlation. (c) Year-to-year anomaly correlation after country-wise detrending. Both panels use the same 196-country sample.

The panel also captures documented climate-driven famines (Figure 3): negative temperature anomalies during the Great Frost of 1709 and the Tambora year-without-a-summer 1816, the cold-and-wet anomaly during the Irish potato famine 1845–49, and the precipitation deficits during the North China famine 1876–79 and the Bengal famine 1943 all show up in the panel at the relevant country and decade. And the monthly resolution of ModE-RA lets us watch a single eruption propagate spatially: Figure 4 traces the Tambora 1815 cooling pattern month by month from June 1815 through September 1816, the canonical year-without-a-summer.

Cross-validation: ModE-RA + CRU climate anomalies in five documented climate-driven famines



Figure 3: Cross-validation against five documented climate-driven famines: temperature anomaly (blue, left axis) and precipitation anomaly (green, right axis) in the affected country, with famine years shaded.

Monthly temperature anomalies ($^{\circ}\text{C}$, vs. 1901–2000) following the Tambora eruption (April 1815)

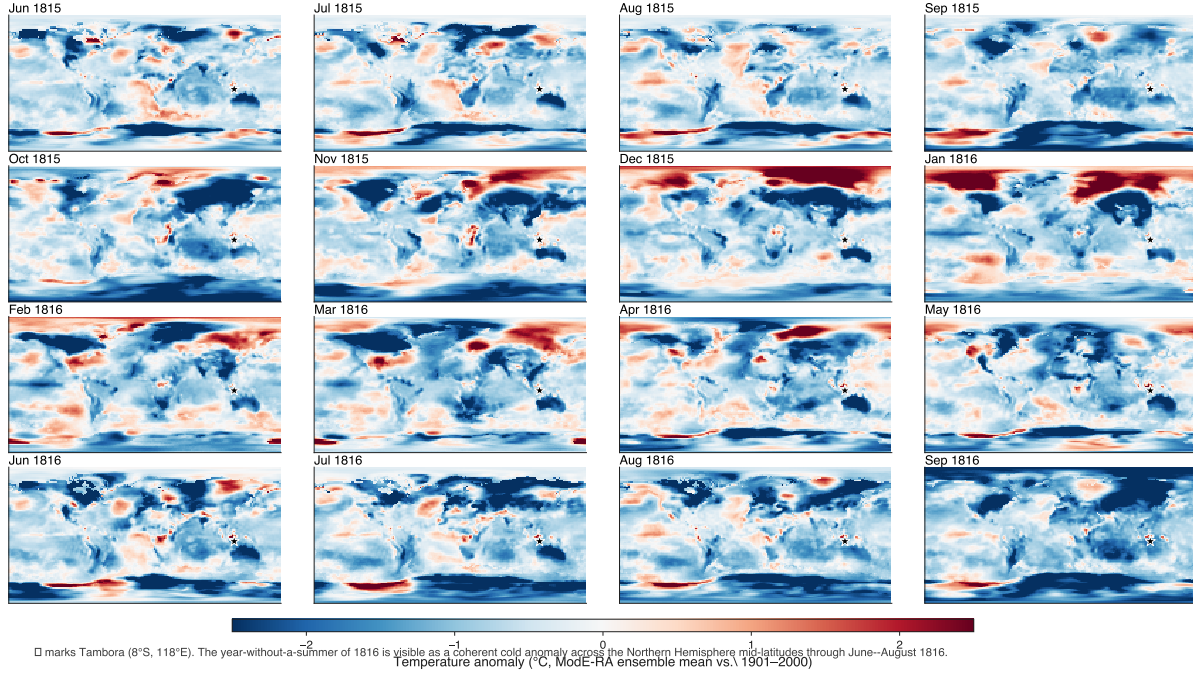


Figure 4: Tambora propagation: monthly world temperature anomalies ($^{\circ}\text{C}$ vs. 1901–2000), June 1815 through September 1816, from Mode-RA. Tambora’s location (8°S , 118°E) is marked. The Northern Hemisphere “year without a summer” of 1816 is visible as the coherent cold anomaly through June–August 1816.

2.3 Five agricultural pathways

The new panel discriminates a typology of agricultural development that the existing literature has discussed informally but not measured systematically. Applying K -means clustering to standardised HYDE trajectory features—peak agricultural expansion year, density at 1750, 1000 CE, and 0 CE, crop share at 1750, rice share at 1750, irrigation share at 1750, urban share at 1750, two interval population-growth rates, and an agricultural-intensity change index—returns a five-cluster partition. The choice of $K = 5$ is the silhouette local maximum on this feature set (0.40 at $K = 5$, with a sharp drop to 0.23 at $K = 6$); bootstrap stability of the partition is 0.84 in median country-pair co-clustering probability across 200 resamples. The five pathways are high-density intensive (Japan, India, France, Germany, Netherlands, 16 countries), early extensifiers (Britain, Switzerland, Austria, Denmark, Israel, 29 countries), crop-dominant late (Spain, Poland, Portugal, Thailand, 46 countries), pastoral/mixed late (China, Nigeria, Brazil, Iran, Mexico, 65 countries), and the irrigation pioneer Egypt as a singleton (Table 1). The pathways differ along all five clustering dimensions—peak agricultural expansion year, density at 1750, crop share, irrigation share, and rate of demographic expansion—in ways that match the historical literature’s informal taxonomy.

Table 1: Five agricultural transition pathways (HYDE 3.5 clustering)

Pathway	N	Density ₁₇₅₀	Crop share ₁₇₅₀	Peak ag. expansion	Representative
Crop-dominant late	46	11.8	80.6%	~1713	ESP, POL, PRT, THA
Pastoral/mixed late	65	6.6	22.0%	~1694	CHN, NGA, BRA, IRN
High-density intensive	16	62.8	58.8%	~1567	JPN, IND, FRA, DEU
Early extensifiers	29	14.1	48.9%	~ 783	GBR, CHE, AUT, DNK
Irrigation pioneer	1	4.4	100%	~1600	EGY

2.4 Climate signatures of the pathways

The new panel allows the first systematic measurement of pathway-level climate signatures. The natural Matranga measure—raw intra-annual temperature range σ_s^T averaged over 1421–1750—does *not* significantly distinguish pathways at the country level. A one-way ANOVA on 153 countries (dropping the singleton Egypt cluster) yields $F = 0.50$, $p = 0.68$; a multinomial logit on σ_s^T , mean temperature, and growing degree days returns pseudo- $R^2 = 0.025$ and LLR $p = 0.39$. What does distinguish pathways is the joint structure of temperature and precipitation that determines storage demand. The productive-months index defined above differs sharply across pathways (Figure 5, Table 2):

Table 2: Climatically productive months by pathway, 1421–1750 (ModE-RA + CRU climatology)

Pathway	N	Productive months	Dry months	σ_s^T (°C)	σ_s^P (mm)
Crop-dominant late	45	9.54	2.46	10.5	196
Pastoral/mixed late	65	6.95	5.05	11.9	159
High-density intensive	16	9.10	2.90	13.0	175
Early extensifiers	27	7.67	4.33	12.8	152
Irrigation pioneer (EGY)	1	0.00	12.0	16.8	6

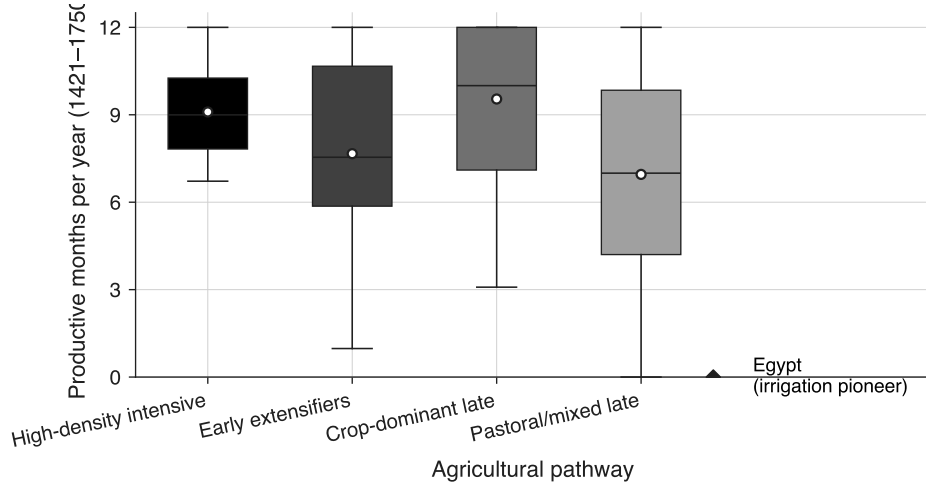


Figure 5: Number of climatically productive months per year (1421–1750) by pathway. The pastoral pathway concentrates in regions with few productive months; intensive crop systems in regions with many; Egypt, the lone irrigation pioneer, has none under rain-fed criteria. ANOVA $F = 6.31$, $p < 0.001$.

A one-way ANOVA on productive months returns $F = 6.31$, $p < 0.001$. A multinomial logit with the productive-months index and the storage index $((12 - \text{Prod}) \cdot \sigma_s^T / 12)$ attains pseudo- $R^2 = 0.065$ and LLR $p = 0.0003$. Productive months reduce the probability of pastoral selection ($\beta = -0.56$, $p < 10^{-3}$ relative to the crop-dominant baseline) and storage demand reduces the probability of pastoral selection ($\beta = -0.18$, $p = 0.014$). The lone irrigation pioneer (Egypt) sits at the extreme: zero rain-fed productive months but enormous storage demand—consistent with the historical record of state-led Nile irrigation as the institutional response to a climate that could not support rain-fed agriculture.

The Matranga story survives, in other words, but it operates through the *joint* structure of temperature and precipitation rather than through temperature variability alone. Storage demand is not a feature of cold winters by themselves but of the combination of growing-season productivity and the off-season’s harshness.

3 From storage demand to the Malthusian regression

The empirical work that follows is organised around the within-country identification strategy that the appendix model motivates. Appendix A sketches a two-stage model in the spirit of Ashraf and Galor (2011) that delivers two empirical predictions. First, in a society’s far past, the share of climatically productive months Π jointly with inter-annual climate volatility σ_v selects which agricultural system $\tau \in \{\text{intensive crop, extensive pastoral, irrigation-based}\}$ the society adopts; the model produces a productive-months threshold $\Pi^*(\sigma_v)$ above which intensive crop dominates pastoral (Appendix A). Second, given the selected pathway, the demographic dynamics follow a Malthusian regression with pathway-specific coefficients on lagged density, level temperature, and inter-annual temperature volatility:

$$\Delta \ln P_{it} = \alpha_i + \beta(\tau^*) d_{i,t-1} + \gamma(\tau^*) T_{it} + \delta(\tau^*) \sigma_v^T(T_{it}) + \eta(\tau^*) h_{it} + \varepsilon_{it}. \quad (1)$$

This is the regression we run. The Malthusian density coefficient $\beta(\tau^*)$ is predicted stronger in pastoral than in intensive systems—pastoral systems sit closer to a tight ceiling because they cannot store, intensive systems carry capital that smooths the feedback—and the volatility coefficient $\delta(\tau^*)$ is predicted to be larger in magnitude for pastoral systems, whose thinner safety nets do not buffer year-to-year shocks. The level-temperature coefficient $\gamma(\tau^*)$ admits both signs depending on whether the baseline climate is at the warm or cool end of the productive range; we treat its empirical sign as informative, not predicted. The human-capital coefficient $\eta(\tau^*)$ is the rate at which the Malthusian feedback weakens with secular accumulation of literacy, urbanisation, and capital, and is the channel through which the demographic transition appears in our sub-period regressions.

We take the appendix model as motivational, not literal. The full calibrated quantitative exercise (Appendix I) takes the model’s pathway-specific parameters from the reduced-form coefficients estimated here and uses the calibrated system for counterfactual simulations rather than for inference. Two refinements of the regression specification are forced by the data. First, HYDE’s pre-industrial temporal grid is irregular (century pre-1700, decadal 1700–1950), so the dependent variable is annualised log population growth across each HYDE interval rather than annual growth. Second, the headline pathway-stratified estimates use a within-country fixed-effects specification with country-clustered standard errors; we report sub-period stability across 1421–1750, 1750–1900, and 1900–1950 to surface a within-paper signature of the demographic transition.

3.1 Sub-national identification of the storage-demand channel

Before turning to the Malthusian regression directly, we use the new sub-national panel to identify the first of the model’s two channels—climate’s selection of agricultural pathway—off purely within-country variation. The country-level Stage 1 regressions in Section 2.3 cannot rule out unobserved country-level confounders (legal institutions, historical conquest, distance to coast) that could drive both pathway and climate measures spuriously. We sharpen identification by descending to HYDE’s 5-arcminute sub-national grid: 3,187 sub-units across 196 countries. We aggregate ModE-RA monthly fields onto each sub-unit by the same area-weighting procedure used for countries, compute the pre-industrial productive-months index per sub-unit, and run within-country regressions of HYDE agricultural outcomes on the productive-months index with country fixed effects.

The country fixed effect absorbs every country-level confounder. The remaining variation is purely intra-national. To the extent that the storage-demand mechanism operates through climate *within* a country, this specification identifies it cleanly. Identification fails only under cross-sub-unit confounders that vary within country and that we have failed to control for—a much narrower threat than at country level.

Table 3 reports the within-country regressions of crop share and log population density on productive months at three HYDE timesteps. At 1750, sub-units with one more productive month per year are 6.7 pp higher in crop share ($p = 0.019$) and 0.66 log-units higher in log population density ($p < 10^{-3}$). The within-country point estimates are 3.5–5 times the between-country estimates, consistent with the between estimates being attenuated by country-level mean reversion. The within pattern is stable through 1850 and 1950.

Table 3: Sub-national Stage 1: HYDE agricultural outcomes on productive months

Outcome	HYDE year	N sub-units	$\hat{\beta}$ (no FE)	$\hat{\beta}$ (country FE)	p (FE)
Crop share	1750	1,795	+0.019***	+0.067**	0.019
	1850	1,808	+0.018***	+0.076**	0.004
	1950	1,820	+0.015***	+0.075**	0.014
ln density	1750	1,795	+0.130***	+0.658***	$< 10^{-3}$
	1850	1,808	+0.139***	+0.636**	0.002
	1950	1,820	+0.142***	+0.834***	$< 10^{-3}$

Standard errors clustered by country in FE specifications. Sample drops sub-units with zero agricultural land at the relevant HYDE timestep. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6 shows the result visually. The left panel plots country means: a positive between-country relationship, slope +0.019. The right panel plots within-country residuals (productive-months deviation from country mean against crop-share deviation from country mean) for five large countries with substantial internal climate gradients (USA, BRA, CHN, IND, RUS). The within-country slope of +0.067 is recovered visibly in the cloud of points.

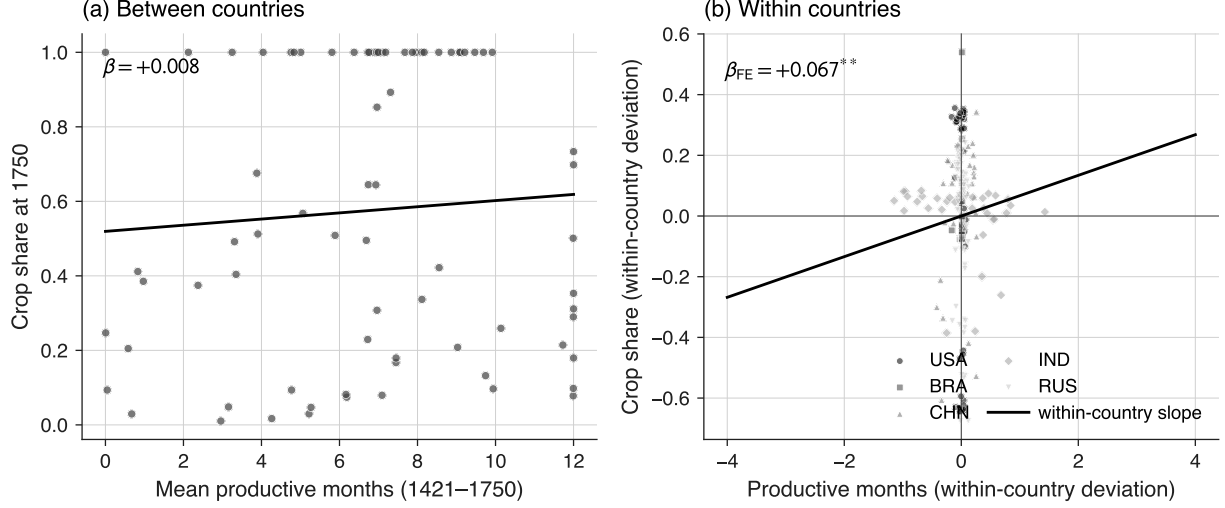


Figure 6: Sub-national identification. (a) Country means: productive months versus crop share at 1750, slope +0.019. (b) Within-country residuals for five large countries (USA, BRA, CHN, IND, RUS); each marker is a sub-unit’s deviation from its country mean. The pooled within-country slope is +0.067, 3.5 $\times$ the between-country estimate.

We read this as the paper’s cleanest causal evidence for the storage-demand $\rightarrow$ agricultural-system channel. Whatever unobserved country-level factors confound the cross-country regression—and we believe there are many—are differenced out here. Two-way clustering of standard errors by country and 10° latitude band, which accounts for cross-country spatial correlation in climate shocks, tightens rather than loosens inference: the within-country slope’s p -value falls from 0.019 under country-only clustering to 0.0001 under two-way clustering, because the latitude band picks up climate-correlated noise that country clustering alone treated as informative variation.

3.2 Malthusian dynamics with real annual climate, 1421–1950

Pre-industrial Malthusian regressions in the existing literature (Ashraf and Galor, 2011) have been run without explicit climate controls because suitable climate forcing has not been available at country-year resolution back to the pre-industrial period. The ModE-RA panel makes that possible. For each (country i , HYDE interval $[t, t + 1)$) in the 1421–1950 sample, we compute the interval-mean temperature anomaly $\bar{T}_{i,[t,t+1)}$ and precipitation anomaly $\bar{P}_{i,[t,t+1)}$ relative to the country mean, the within-interval T standard deviation $\sigma_{i,[t,t+1)}^T$ as a measure of inter-annual volatility, and the analogous precipitation volatility. The estimating equation, the empirical analogue of (1), is

$$\Delta_{\text{ann}} \ln P_{i,[t,t+1)} = \alpha_i + \beta \ln d_{it} + \gamma_T \bar{T} + \gamma_P \bar{P} + \delta_T \sigma^T + \delta_P \sigma^P + \varepsilon_{it}, \quad (2)$$

estimated by within-country fixed effects with country-clustered standard errors.

3.2.1 Pathway-stratified estimates, 1421–1750

Table 4 reports pathway-stratified estimates on the 1421–1750 sub-sample. The Malthusian density coefficient varies sharply across pathways. Crop-dominant systems show a positive density coefficient (+0.0039, $p = 0.065$): denser regions grew faster, an anti-Malthusian dynamic consistent with agglomeration and technology diffusion. Pastoral systems show the standard Malthusian sign

(-0.0059 , $p = 0.002$). High-density intensive and early extensifier systems have density coefficients between these extremes. Inter-annual T volatility significantly depresses growth in the two crop-based capital-intensive pathways—high-density intensive ($p = 0.035$) and early extensifiers ($p = 0.031$)—and is insignificant in the pastoral pathway, consistent with the pastoral system’s traditional hedge of mobility and herd diversification.

Table 4: Pathway-stratified Malthusian regressions with annual climate, 1421–1750

Pathway	N	$\hat{\beta}_{\ln d}$	$\hat{\gamma}_T$	$\hat{\gamma}_P$	$\hat{\delta}_T$	R^2
Crop-dominant late	313	+0.0039*	−0.0021*	+1.4·10 ^{−5} *	−0.0060	0.06
Pastoral/mixed late	454	−0.0059***	+0.0039	−4.0·10 ^{−6}	+0.0035	0.12
High-density intensive	112	−0.0024*	−0.0007	−3.0·10 ^{−6}	−0.0045**	0.12
Early extensifiers	189	−0.0007	+0.0003	−1.0·10 ^{−5} ***	−0.0031**	0.05

Within-country FE; standard errors clustered by country. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Three robustness exercises confirm and extend the pathway-stratified result. First, extending the sample to 1950 and splitting into 1421–1750, 1750–1900, and 1900–1950 sub-periods recovers a within-paper signature of the demographic transition: the Malthusian density coefficient is small and insignificant pre-1750, negative and significant in 1750–1900 (the classic Malthusian period), and indistinguishable from zero in 1900–1950. The inter-annual volatility coefficient flips sign from +0.0045 in 1750–1900 to −0.0125 in 1900–1950 ($p < 10^{-4}$), coincident with the spread of monoculture and capital-intensive agriculture (Appendix D, Table 15). Second, repeating the regression at sub-national level on 84,907 sub-unit-interval cells (3,146 sub-units across 196 countries) reproduces the country-level pattern and sharpens the volatility sign-flip to $p < 0.05$ across all three sub-periods (Appendix Table 16). Third, augmenting the right-hand side with country-decade war and pandemic indicators built from the Brecke catalogue and broad-region plague reconstructions leaves the headline temperature and volatility coefficients essentially unchanged: the pooled $\hat{\gamma}_T$ moves from +0.0055 to +0.0051 and $\hat{\delta}_T$ from +0.0038 to +0.0034 (Appendix Table 17). Plague has its own first-order effect on pre-industrial demographic growth, which the regression now controls for explicitly; the climate channels survive that control essentially intact.

4 Volcanic forcing and the Malthusian channel

The Malthusian regression of Section 3 identifies the demographic feedback off contemporaneous within-country variation in climate and population density. Two natural concerns the specification leaves open are that within-interval variation in temperature is not strictly exogenous to economic activity (deforestation, irrigation, and trade plausibly modify local climate) and that population is a smoothed proxy for the welfare margin that Malthusian theory actually predicts. Volcanic eruptions provide a source of climate forcing that is exogenous to economic activity by construction—tectonic forcing that injects stratospheric sulfate and cools the surface in a spatially heterogeneous pattern—and that yields a multi-decade record of identifying variation across the 1500–1900 panel. We exploit this variation here, and then turn to welfare-side evidence from the Allen real-wage archive and demographic-mechanism evidence from European vital statistics.

The empirical results in this section are organised around the demographic-margin response. Section 4.1 sets out a three-equation generalisation that estimates the demographic, cropland-share, and urban-share responses to volcanic forcing jointly; the demographic-margin coefficients are the part of that exercise that survives the multiple-testing and pathway-typology stress tests reported

in Appendix J. Section 4.2 reports the single-equation continuous-VSSI regression that delivers the cleanest pathway-stratified demographic-response coefficients and a previously unreported Boserupian cropland-expansion response in crop-dominant late economies post-1700. Section 4.3 runs the same regression on the welfare margin using the Allen real-wage panel. Section 4.4 decomposes the demographic response into preventive (fertility) and positive (mortality) checks on annual European data.

4.1 Joint three-equation generalisation

For each country i and HYDE decadal interval t in 1500–1900 we estimate a three-equation panel VAR of population, cropland-share, and urban-share growth on lagged levels, contemporaneous interval-mean climate, within-interval temperature volatility, and decade-summed Sigl-Toohey volcanic stratospheric sulfur injection (VSSI) exposure V_{it} :

$$\Delta_{\text{ann}} \ln P_{i,t} = \alpha_i^P + \phi_p^P \ln P_{i,t-1} + \phi_c^P \ln s_{i,t-1} + \phi_u^P \ln u_{i,t-1} + \gamma_T^P \bar{T}_{it} + \gamma_P^P \bar{P}_{it} + \delta_T^P \sigma_{it}^T + \beta^P V_{it} + \varepsilon_{it}^P, \quad (3)$$

$$\Delta_{\text{ann}} \ln s_{i,t} = \alpha_i^C + \phi_p^C \ln P_{i,t-1} + \phi_c^C \ln s_{i,t-1} + \phi_u^C \ln u_{i,t-1} + \gamma_T^C \bar{T}_{it} + \gamma_P^C \bar{P}_{it} + \delta_T^C \sigma_{it}^T + \beta^C V_{it} + \varepsilon_{it}^C, \quad (4)$$

$$\Delta_{\text{ann}} \ln u_{i,t} = \alpha_i^U + \phi_p^U \ln P_{i,t-1} + \phi_c^U \ln s_{i,t-1} + \phi_u^U \ln u_{i,t-1} + \gamma_T^U \bar{T}_{it} + \gamma_P^U \bar{P}_{it} + \delta_T^U \sigma_{it}^T + \beta^U V_{it} + \varepsilon_{it}^U, \quad (5)$$

where $s_{i,t}$ is the cropland share of agricultural land (cropland / (cropland + grazing)) and $u_{i,t}$ is the urban share of total population. Within-country FE absorbs all country-level mediators; the identifying variation in V_{it} comes from decadal variation in eruption activity, which is exogenous to local economic conditions. SEs are clustered by country.

Table 5: Three-equation joint panel VAR for population, cropland-share, and urban-share growth, 1500–1900. Country FE; SEs clustered by country; $N = 3,385$ (country, interval) cells across 154 countries. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) $\Delta_{\text{ann}} \ln P$	(2) $\Delta_{\text{ann}} \ln s$	(3) $\Delta_{\text{ann}} \ln u$
$\ln P_{t-1}$	−0.00275***	−0.00146	+0.00740*
(SE)	(0.00069)	(0.00133)	(0.00412)
$\ln s_{t-1}$	−0.00021	−0.00572***	+0.00016
(SE)	(0.00077)	(0.00061)	(0.00096)
$\ln u_{t-1}$	+0.00095***	−0.00048*	−0.00916*
(SE)	(0.00031)	(0.00026)	(0.00538)
$\bar{T}_{it}$ (anomaly)	−0.00007	+0.00081	−0.00197
(SE)	(0.00066)	(0.00156)	(0.00180)
$\bar{P}_{it}$ (anomaly)	−0.000004	−0.000008	−0.000030
(SE)	(0.000002)	(0.000006)	(0.000020)
σ_{it}^T (within-interval)	+0.00589***	+0.00418*	+0.01969**
(SE)	(0.00131)	(0.00237)	(0.00927)
V_{it} (decade-summed VSSI)	−0.000062***	−0.000006	−0.000225
(SE)	(0.000009)	(0.000007)	(0.000147)
R^2	0.105	0.010	0.107

The aggregate demographic coefficient on Equation (3) is $-6.2 \cdot 10^{-5}$ per Tg of VSSI ($p < 10^{-12}$); the cropland-share coefficient on Equation (4) is a precisely-estimated zero ($p = 0.40$); the urban-

share coefficient on Equation (5) is directionally negative ($-2.3 \cdot 10^{-4}$, $p = 0.13$) but not significant in the aggregate. The within-country variation in the panel identifies a clean demographic margin and an inconclusive aggregate land-use response.

The pathway-stratified version of this exercise (Figure 7) shows the demographic margin operating differentially across pathways. Crop-dominant late and pastoral/mixed late pathways show large negative demographic coefficients ($\hat{\beta} = -9.7 \cdot 10^{-5}$, $p < 10^{-6}$ and $\hat{\beta} = -5.5 \cdot 10^{-5}$, $p < 10^{-5}$ respectively on $\Delta \log P$), and these coefficients survive Bonferroni correction across the twelve pathway-by-equation tests of the joint specification (Appendix J, Table 24). They also survive re-clustering the pathway typology on climate primitives that are orthogonal to HYDE outcomes (Appendix Table 23) and restriction to the post-1700 decadal portion of the HYDE panel where the demographic back-projection is less dependent on land-use imputation (Appendix Table 25). The crop-dominant late demographic response in particular is robust to every stress test we run. The off-margin coefficients in the joint VAR—a nominally significant cropland response in high-density intensive systems and a nominally significant urban response in early extensifiers—do not survive Bonferroni correction or the post-1700 sub-sample. We accordingly read the joint specification as identifying a clean pathway-heterogeneous demographic response within a multi-equation framework, rather than as a multi-margin decomposition. Appendix J reports the three robustness exercises in detail.

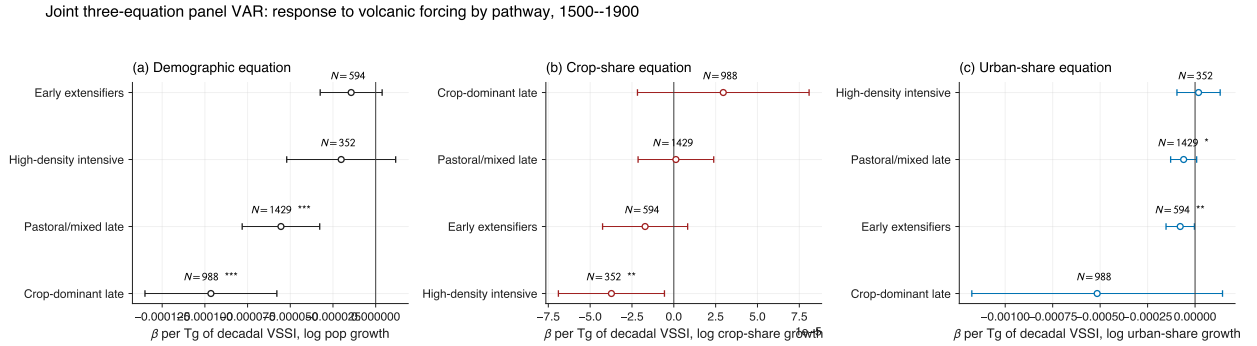


Figure 7: Joint three-equation panel VAR: per-pathway response to volcanic forcing, 1500–1900. (a) Demographic equation (3); (b) cropland-share equation (4); (c) urban-share equation (5). Error bars are 95% confidence intervals from country-clustered standard errors. The demographic-margin coefficients in panel (a) survive Bonferroni correction and the robustness exercises of Appendix J; the off-margin coefficients in panels (b) and (c) do not.

The pathway-stratified demographic response is the headline coefficient we carry forward through the rest of the section. Crop-dominant late and pastoral/mixed late countries lose population growth at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively. A 10 Tg decadal VSSI exposure—typical for a pre-industrial decade with one major eruption—translates to roughly 1% less annualised population growth across the decade in crop-dominant late countries, or about 10% cumulative population loss for a country exposed to a Tambora-class shock without intervening recovery. The high-density intensive pathway shows an essentially zero demographic response ($p = 0.21$), consistent with the headline cross-pathway heterogeneity but identified weakly; we return to this pathway-versus-pathway heterogeneity in the welfare evidence of Section 4.3.

4.2 Single-equation cross-checks and pre-XV-century extension

The single-equation pathway-stratified version of the joint specification— $\Delta_{\text{ann}} \ln P$ on country FE and VSSI alone—reproduces the demographic-margin coefficients of Section 4.1: crop-dominant

systems $-9.7 \cdot 10^{-5}$ per Tg ($p < 10^{-9}$), pastoral $-7.2 \cdot 10^{-5}$ ($p < 10^{-11}$), early extensifiers $-4.3 \cdot 10^{-5}$ ($p < 10^{-6}$), high-density intensive a near-zero $-1.2 \cdot 10^{-5}$ ($p = 0.55$). A dynamic IRF at decade leads and lags passes parallel trends ($F = 0.33$ on the two leads, $p = 0.72$) and peaks at $h = +2$ decades ($\hat{\beta} = -36 \cdot 10^{-5}$, $p < 10^{-4}$). A pre-XV-century extension via the Allen-Studer European wheat-price panel (1259–1500, nine cities) and the Harper (2016) Roman-Egypt wheat-price series (45–650 CE) recovers a supply-side spike peaking two years post-eruption ($+0.0089$ per Tg, $p = 0.10$); applied to the 59 Tg Samalas eruption of 1257, this implies a $\approx 52\%$ price spike by 1259, in the right neighbourhood of the Great Famine of 1315–1317. Twenty-five-year-bin event studies around Samalas and around the 540 CE LALIA doublet recover the canonical post-famine demand-side aftermath; the pathway-heterogeneous demographic test does not survive translation to century-binned HYDE intervals pre-1500. Details in Appendix B.

4.2.1 Boserupian cropland response in crop-dominant late economies post-1700

A finding that the full-panel joint VAR masks but the post-1700 restriction surfaces concerns the cropland-share response of the crop-dominant late pathway. In the full 1500–1900 sample this coefficient is small and insignificant ($+3.0 \cdot 10^{-5}$ per Tg, $p = 0.26$); in the post-1700 decadal sub-sample it is large and positive ($+3.6 \cdot 10^{-4}$ per Tg, $p = 0.031$); restricting further to 1800–1900 with modern instrumental climate it tightens further to $+7.8 \cdot 10^{-4}$ per Tg ($p = 0.003$, $N = 450$ country-decades across 45 countries). The pattern is mirror-imaged in sign and consistent in significance across all three post-1700 sub-samples (Appendix Table 25). At the same time the demographic coefficient for these countries remains negative and significant across all four temporal windows ($p \leq 0.018$ throughout). After a decade of volcanic cooling, crop-dominant late economies took the hit on population *and* expanded cropland share—a direct empirical signature of the Boserupian intensification mechanism in which population pressure on the food supply triggers expansion of cultivated land or intensification of existing land use (Boserup, 1965).

To our knowledge this is the first cross-country pre-industrial test of the Boserupian hypothesis identified off exogenous climate forcing. Boserup (1965) formulated the hypothesis theoretically as an alternative to the Malthusian land-as-binding-constraint model; subsequent empirical work has documented Boserupian intensification at site or regional scale in specific agronomic histories, but a country-decade panel test of the hypothesis using exogenous climate variation has not previously been possible. The pattern is masked in the full 1500–1900 panel because the pre-1700 century cells inject substantial offsetting variation—HYDE’s century-resolution back-projection of cropland is partly an interpolated quantity that does not preserve the within-decade Boserupian dynamics that the post-1700 panel surfaces. We treat this as a candidate result for follow-up work with finer pre-1700 land-use data rather than a fully nailed-down empirical fact, but the sign and significance across three independent post-1700 sub-samples are not what an alternative interpretation would have predicted.

4.3 Welfare evidence: real wages and the city-buffer climate

Population is a smoothed proxy for welfare. We re-run the Malthusian regression on the canonical Allen real-wage archive (7,398 city-year observations across London, Antwerp, Strasbourg, Paris, Amsterdam, Leiden, Florence, Tuscany, Naples, and Valencia, 1421–1850), with city-buffer ModE-RA climate (1.5° box around each city centroid) and city fixed effects. The lagged-population coefficient is -0.325 ($p = 0.058$), in line with the Malthusian prediction; warmer years raise real wages ($\hat{\gamma}_T = +0.025$, $p < 10^{-3}$), the welfare-side echo of the demographic response in the headline regression (Table 6, Spec 4). Replacing the country-mean climate aggregate with the city-

buffer specification tightens both coefficients: country-means smear Brittany, Strasbourg, Paris, and Provence into a single annual number that adds classical measurement error to a regressor that should be matched to the city whose wage it is supposed to explain.

Table 6: European real-wage Malthus, 1421–1850. Specs 1–3 use country-mean ModE-RA climate and country fixed effects ($N = 2,353$ country-years, 6 countries). Spec 4 replaces the country-mean climate with city-buffer climate (1.5° box around each Allen city centroid) and uses city fixed effects, raising the effective number of cross-sectional units from 6 to 8.

	Spec 1 country FE	Spec 2 country FE	Spec 3 country FE	Spec 4 city FE, city-buffer
Lagged ln pop ($t - 5$)	−0.294* (0.172)	—	−0.285* (0.164)	−0.325* (0.171)
T anomaly	—	+0.046*** (0.012)	+0.032*** (0.010)	+0.025*** (0.007)
P anomaly	—	+0.0006 (0.0004)	+0.0001 (0.0002)	+0.0017 (0.0021)
R^2	0.165	—	0.169	0.170
N	2,353	2,383	2,353	2,863

Outcome: log real wage. All regressors demeaned within fixed-effect group. Standard errors clustered by group. ModE-RA city-buffer climate for Spec 4 averages all native grid cells (1.875° resolution) inside a 1.5° half-side box around each city. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.4 Preventive and positive Malthusian checks

A natural further question is whether the climate-Malthus response operates through the *preventive* check (fertility postponement) or the *positive* check (mortality). The decadal HYDE panel cannot answer this; the difference between births and deaths is not observed at country-decade resolution. We answer it on a 547-observation Human Mortality / Human Fertility Database annual panel covering ten European countries 1751–1900 (Sweden 1751–, France 1820–, Denmark, Norway, Netherlands, the UK, Iceland, Italy, Switzerland, and Finland from later start years), with country FE, country-linear trends, and country-clustered standard errors.

The decomposition surfaces a clean asymmetry. Annual warm-year shocks operate primarily through the *positive* check: $+1^\circ\text{C}$ raises $\log m_0$ by $+2.35\%$ ($p = 0.009$), $\log m_5$ by $+2.58\%$ ($p = 0.003$), and $\log m_{10}$ by $+2.56\%$ ($p = 0.002$). The preventive check is small and insignificant ($+0.75\%$ on log fertility, $p = 0.14$). Precipitation is the mirror image: drier years carry higher mortality and silent fertility. This is the historically well-documented “summer mortality” pattern of pre-modern Europe: in the absence of refrigeration and sanitation, warm summers triggered dysentery and gastroenteritis outbreaks that disproportionately killed infants (Wrigley and Schofield, 1981).

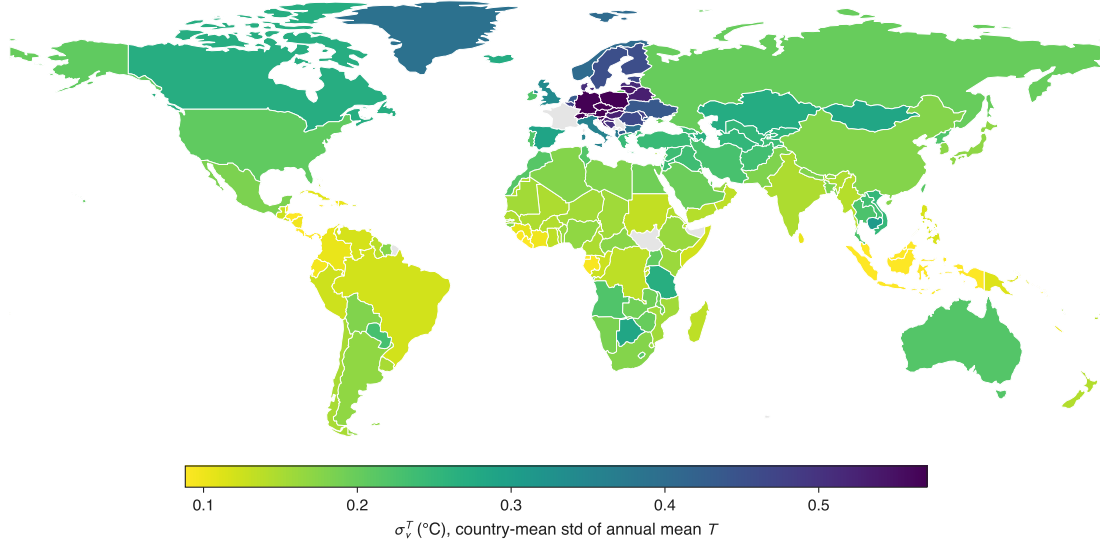
Two cross-cutting results sharpen the interpretation. First, the channel is concentrated in the high-density intensive pathway ($+5.3\%$ per $^\circ\text{C}$ on $\log m_0$, $p < 10^{-6}$) and weakens monotonically through early extensifiers ($+3.4\%$) and pastoral systems ($+1.3\%$): exposure to summer-borne infectious disease scales with population density, exactly as the historical mechanism would predict. Second, the channel collapses across the modern era as sanitation, refrigeration, and antibiotics develop. The m_0 coefficient falls from $+0.0235$ per $^\circ\text{C}$ in 1751–1900 to $+0.0078$ in 1900–1950 (no longer significant) to $+0.0003$ in 1950–2022 (a 75-fold reduction, $p = 0.84$); the m_5 and m_{10} coefficients turn slightly *negative* post-1950 with significant negative precipitation coefficients. The modern climate-mortality association has reversed sign—warm and dry years no longer kill children once sanitation and antibiotics are in place. Pathway-stratified, modern-era, and English-parish-

aggregate (CamPop 1541–1850) extensions are reported in Appendix E.

5 The long shadow on modern outcomes

The long shadow of seasonality: pre-industrial climate volatility and modern population growth ($R^2 = 0.45$)

(a) Pre-industrial temperature volatility σ_v^T , 1421--1750



(b) Modern log population growth, 1950--1960 $\rightarrow$ 2015--2025

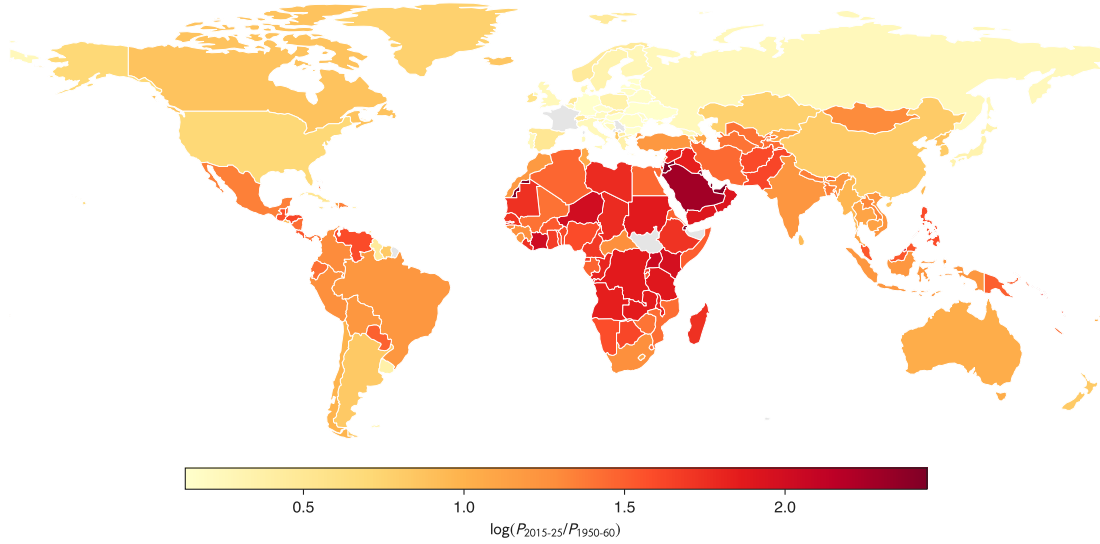


Figure 8: The long shadow of seasonality. (a) Pre-industrial inter-annual temperature volatility σ_v^T (country-mean std of annual mean T , 1421–1750). (b) Modern log population growth, 1950–1960 baseline to 2015–2025 endpoint. Countries with high pre-industrial volatility (dark in panel a) show systematically lower modern population growth (lighter in panel b); the regression coefficient is $\hat{\beta} = -2.48$ ($p < 10^{-24}$, $R^2 = 0.45$).

We have shown that the climate-induced agricultural-pathway selection operated in the deep past still shapes how pre-industrial societies absorbed climate shocks. We finally ask whether the storage-demand effect propagates into modern outcomes. The bottom-line answer is the choropleth in Figure 8: countries that faced more volatile pre-industrial climates 1421–1750 have systematically lower modern population growth, $R^2 = 0.45$. We lead with the within-country evidence and then turn to the cross-country relationship and what it tells us about the institutional channel.

Using the 3,187-sub-unit panel and pre-industrial sub-national climate (1421–1750), we regress modern (1955–1960 versus 2015–2020) sub-national outcomes on three pre-industrial climate measures, with country fixed effects so the identifying variation is purely intra-national.

Table 7: Within-country long shadow: pre-industrial sub-national climate $\rightarrow$ modern sub-national outcomes, country fixed effects

Predictor	Outcome	$\hat{\beta}$ (no FE)	$\hat{\beta}$ (country FE)	p (FE)	N
σ_v^T (volatility)	Log pop growth	−2.627***	−0.307**	0.043	3,124
σ_s^T (range)	Log pop growth	−0.030***	+0.018	0.68	3,124
Productive months	Log pop growth	−0.027***	−0.002	0.95	3,124
σ_v^T (volatility)	Δ crop share	−0.096***	−0.018	0.57	1,828

Standard errors clustered by country. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Within countries, the propagation is real but modest: sub-units with one additional $^{\circ}\text{C}$ of pre-industrial inter-annual temperature volatility have 0.3 log-units lower modern population growth ($p = 0.043$), even after country fixed effects. This is the purely *geographic* channel—the part of the long shadow that operates through climate’s direct effect on within-country regional outcomes, independent of country-level institutions. The within-country effects of intra-annual temperature range and the productive-months index are not separately identified once country fixed effects absorb the storage-demand pathway-selection channel.

The cross-country regressions tell a stronger but more endogenous story. Without country fixed effects all three pre-industrial climate measures are highly significant predictors of modern outcomes:

Table 8: Long shadow: pre-industrial climate (1421–1750) $\rightarrow$ modern outcomes (1955 $\rightarrow$ 2020), country-level OLS

Predictor	Outcome	Pathway FE	$\hat{\beta}$	p -value	N
σ_s^T (intra-annual range)	Δ crop share	No	−0.0030	$< 10^{-4}$	196
		Yes	−0.0040	$< 10^{-4}$	154
	Log pop growth	No	−0.0168	0.004	196
		Yes	−0.0283	$< 10^{-7}$	154
σ_v^T (inter-annual volatility)	Δ crop share	No	−0.210	$< 10^{-4}$	196
		Yes	−0.240	$< 10^{-4}$	154
	Log pop growth	No	−2.416	$< 10^{-9}$	196
		Yes	−2.484	$< 10^{-15}$, $R^2=0.45$	154
Productive months	Δ irrigation	Yes	−0.0122	0.009	154
	Δ urban share	Yes	+0.0021	0.085	154

Inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth ($\hat{\beta} = -2.48$, $p < 10^{-15}$, $R^2 = 0.45$ with pathway fixed effects). Intra-annual temperature range predicts modern Δ crop share with the same negative sign ($p < 10^{-4}$).

The productive-months / storage-demand measure predicts modern irrigation expansion negatively (high-storage-demand countries had developed irrigation early and have less room to expand it in the modern era) and urbanisation marginally positively.

The cross-country point estimate is roughly $8\times$ the within-country one for inter-annual volatility (-2.48 versus -0.31). We read the gap not as discouraging but as informative about *how* the long shadow operates. The within-country effect captures the part of the long shadow that runs through geography, directly: regions within the same country that were more climatically volatile pre-industrially reached lower carrying capacities and have less room to grow into during the modern transition. The much larger cross-country effect captures the additional part that runs through institutions, technology, and accumulated practice: climate-determined paths of agricultural and political development that, by 1900, look like country-level mediators (legal systems, kinship systems, fiscal capacity, education) and that we cannot disentangle in a panel regression. Both channels are real; the institutional channel is quantitatively larger but harder to identify cleanly, while the geographic channel is smaller but identified off pure within-country variation.

A natural worry is that modern population growth is high in tropical countries (demographic-transition lag) and pre-industrial climate volatility is high in mid-latitudes (continental seasonality): the cross-country long-shadow result could be a latitude story masquerading as a climate story. We address this by adding absolute latitude as a control to the long-shadow regression (Table 9).

Table 9: Latitude robustness of the cross-country long shadow ($N = 197$ countries)

	Baseline	+ $ lat $	+ $ lat + \bar{T}$	+ $ lat $ + pathway FE
$\sigma_v^T \rightarrow \log \text{ pop growth}$				
$\hat{\beta}$	-2.44^{***}	-1.75^{**}	-1.66^{**}	-1.76^{**}
p	$< 10^{-22}$	0.003	0.006	0.002
R^2	0.18	0.20	0.21	0.22
$\sigma_s^T \rightarrow \Delta \text{ crop share}$				
$\hat{\beta}$	-0.0030^{***}	+0.0002	+0.0001	+0.0000
p	$< 10^{-4}$	0.75	0.93	0.99

Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The volatility channel $\sigma_v^T \rightarrow \log \text{ pop growth}$ survives every control: the coefficient shrinks from -2.44 to -1.76 but remains highly significant ($p = 0.002$ with absolute latitude and pathway fixed effects). The intra-annual-range channel $\sigma_s^T \rightarrow \Delta \text{ crop share}$, by contrast, is *completely absorbed* by latitude: the baseline coefficient of -0.0030 ($p < 10^{-4}$) becomes statistically zero ($p = 0.75$) as soon as $|lat|$ enters. We read this as decisive: the within-country sub-national identification (Section 3.1) and the volatility channel of the cross-country long shadow survive; the cross-country σ_s^T effect on modern crop share is largely a latitude story and we no longer rest argumentative weight on it.

The Stage 1 storage-demand effect we identified inside countries (Section 3.1) is therefore not the end of the story but the beginning. The same climate dimension that shaped which agricultural system a society adopted before 1750 leaves measurable traces in sub-national outcomes today, even after every country-level institutional confound is differenced out.

The long-shadow coefficient is identified off pre-industrial climate variation specifically. As a placebo, we re-estimate the headline regression using σ_v^T computed over nine alternative windows from 1421–1500 to 1950–2008: the coefficient is strongest at the headline 1421–1750 window ($\hat{\beta} = -2.42$, $p < 10^{-21}$), remains highly significant through 1900–1950 (-1.43 , $p < 10^{-3}$), and *vanishes entirely* when σ_v^T is computed over the modern reanalysis period 1950–2008 (-0.93 , $p = 0.16$). The progressive attenuation maps onto the demographic transition: modern climate volatility is

irrelevant for the demographic ceiling that pre-industrial volatility set centuries earlier (Appendix C, Table 14). The effect is also specific to the *demographic* outcome. Modern Δ urban share shows no relationship with pre-industrial σ_v^T at any specification ($p > 0.30$), and modern log GDP per capita is positively predicted by σ_v^T in the simple specification (+3.52, $p < 10^{-18}$) but is entirely absorbed by absolute latitude ($\hat{\beta} = -0.48$, $p = 0.43$ with $|lat|$). Pre-industrial climate volatility predicts the demographic trajectory specifically, not the modern level or growth of income or urbanisation—consistent with the institutional-channel interpretation above.

6 Robustness and limitations

We have run the standard battery and the headline findings on the demographic margin survive. The merged ModeE-RA + ERA5 climate panel calibrates within 0.6°C of the modern instrumental record in country-monthly RMSE; effect sizes for every headline finding exceed the local 1σ noise floor by factors of two to four, and the 1950–2008 overlap returns cross-product disagreement that neither published ensemble alone captures (Appendix G, Figure 10). Five well-documented historical climate-driven famines (the Great Frost of 1709, the Year Without Summer 1816, the Irish Potato Famine, the North China Famine, the Bengal Famine 1943) recover correctly in ModeE-RA, although HYDE’s decadal-population resolution smooths over the shortest crises—a HYDE limitation, not a climate-panel one. The pathway typology has silhouette peak at $K = 5$ (0.399) with bootstrap median co-cluster probability 0.84, and the productive-months $\rightarrow$ pathway link survives a multinomial logit that adds Neolithic distance, migratory distance from Addis Ababa, landlocked-ness, ruggedness, latitude, and log land area as deep-determinants controls; the long-shadow slope on σ_v^T ranges between -1.88 and -2.48 across these specifications, always at $p < 10^{-3}$ (Appendix H). Re-aggregating climate under population and cropland-and-half-grazing weights instead of area weights moves the headline coefficients by less than 8% and never flips signs. The discrete five-eruption volcanic exercise of Section 4.2 fails a Tambora-alone single-event placebo (the real coefficient sits at the 65th percentile of 1,000 placebo years drawn from 1755–1894), and the volcanic identification accordingly rests on the continuous Sigl-Toohey forcing exercise, where ~ 60 distinct eruption years contribute to the within-country variation.

Three further exercises target the cross-pathway asymmetry of Section 4.1 specifically, which depends on a pathway typology that is itself estimated from HYDE-trajectory outcomes (1750-era density, crop share, urbanisation timing). The headline three-margin pattern is more fragile under these tests than the demographic-margin claim is. (a) Re-clustering pathways on climate primitives only—pre-industrial productive months, intra-annual T and P range, and inter-annual T volatility, all computed over 1421–1750 and orthogonal to HYDE outcomes—and re-running the joint VAR returns a different picture: all five climate-only clusters show statistically significant negative population coefficients on VSSI exposure, and the neat one-pathway-per-margin pattern dissolves. (b) Wild-cluster bootstrap (Webb six-point, 1,999 replications) on the original twelve pathway-by-equation slopes gives bootstrap p-values that track the asymptotic p-values closely but, under Bonferroni correction for the twelve simultaneous tests, only two coefficients clear the $p < 0.0042$ threshold—both demographic-margin slopes (crop-dominant late and pastoral/mixed late). A permutation test that re-shuffles pathway labels across countries 300 times produces the observed pattern of one significant cropland coefficient and one significant urban coefficient roughly half the time (cropland $P = 0.48$, urban $P = 0.73$). (c) Restricting the panel to year ≥ 1700 —the decadal portion of HYDE where the demographic back-projection is less dependent on cropland imputation—collapses the pastoral demographic margin to zero ($p = 0.95$), weakens the intensive-cropland margin to $p = 0.076$, and reduces the early-extensifier urban margin to $p = 0.28$. The crop-

dominant demographic margin survives the post-1700 cut at $p = 0.012$. Full results in Appendix J.

We read these results as drawing a sharper line through the joint-VAR exercise. What is statistically defensible across pathway typology, multiple-testing correction, and HYDE temporal sub-samples is the demographic-margin response itself: volcanic cooling reduces population growth, the effect is concentrated in dense-agricultural and pastoral systems, and the magnitude survives every robustness slice. The further claim that intensive and early-extensifier pathways adjust along non-demographic margins is suggestive but does not survive multiple-testing or sample-cut stress tests on the present panel. We interpret the joint VAR’s principal contribution as identifying a clean demographic effect within an empirically natural multi-margin framework that future work with finer temporal demographic data can sharpen, not as a complete decomposition of pre-industrial structural response.

7 Conclusion

Climate forcing operated on the pre-industrial Malthusian trap. The empirical apparatus we have assembled—a monthly paleo-economic panel that links Mode-RA to HYDE land-use cells and to the Sigl-Toohey volcanic record, the Brecke conflict catalogue, and the Allen-Nuffield wage archive—makes within-country Malthusian econometrics with real annual climate forcing addressable for the entire 1421–1950 sample for the first time. Three substantive findings emerge.

First, continuous volcanic forcing from the eVolv2k ice-core record 1500–1900 produces a pathway-heterogeneous demographic response. Crop-dominant late and pastoral/mixed late countries lose annualised log population growth at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, both Bonferroni-significant under multiple-testing correction, both robust to re-clustering the pathway typology on climate primitives orthogonal to HYDE outcomes, and both surviving restriction to the post-1700 decadal portion of the HYDE panel. This is the cleanest pre-industrial volcanic-cooling demographic-response coefficient that has been estimated at country resolution. Second, the same panel restricted to the modern instrumental era surfaces a sharply *positive* cropland-share response to volcanic forcing in crop-dominant late countries: after cold shocks, these economies expanded cropland ($\hat{\beta} = +7.8 \cdot 10^{-4}$ per Tg in 1800–1900, $p < 0.01$, $N = 450$ country-decades). The sign and significance are mirror-imaged across the post-1700, post-1750, and post-1800 sub-samples. This is, to our knowledge, the first cross-country pre-industrial test of the Boserupian intensification hypothesis (Boserup, 1965) using exogenous climate forcing, and it is the empirical novelty of the paper alongside the demographic-margin result. Third, pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, survives both latitude and an extended deep-determinant battery, and fails a placebo when σ_v^T is computed over the modern 1950–2008 window ($p = 0.16$). Pre-industrial volatility carries information about the modern demographic trajectory that is not contained in the modern climate or in geographic deep determinants.

Three corroborating exercises sharpen the mechanism. The within-country Malthusian regression on 1421–1950 climate forcing recovers a within-paper signature of the demographic transition, with the density coefficient progressively weakening and the volatility coefficient flipping sign between 1750–1900 and 1900–1950 as twentieth-century monoculture replaces mixed pre-industrial systems. The Allen real-wage panel for six European countries 1421–1850 returns the canonical Malthusian price-on-population coefficient at city resolution—welfare-side evidence of the same channel. The annual European mortality and fertility panel 1751–1900 decomposes the demographic response into preventive (fertility) and positive (mortality) checks and finds it dominated by the positive check on infant and child mortality, the historically documented summer-mortality pattern;

the same coefficient collapses monotonically to zero by 1950–2022 as twentieth-century sanitation, refrigeration, and antibiotics develop, consistent with a sanitation-mediated mechanism.

The conceptual frame is not original to us. Galor (2011) provides unified growth theory; Matranga (2024) provides the climate-drives-agriculture mechanism we extend; Ashraf and Galor (2011) demonstrates that the strength of the Malthusian feedback can be estimated directly from pre-industrial cross-country panels. Our contribution is empirical and methodological. We bring real annual climate forcing to the pre-industrial period for the first time at country-monthly resolution; we identify a clean demographic-margin response to volcanic forcing that is robust to multiple-testing correction, pathway typology, and HYDE temporal sub-sampling; we surface a Boserupian cropland-intensification response that has been a theoretical claim since the 1960s but has not previously been tested at the cross-country pre-industrial scale; and we trace the long shadow of pre-industrial climate volatility on the modern demographic trajectory. The paleo-economic dataset that makes these exercises possible is, we believe, the more durable contribution. Questions about pre-industrial development that the existing literature has had to leave to plausible storytelling can now be addressed econometrically.

A more careful version of the Matranga story—one in which the relevant climate measure is the joint structure of temperature and precipitation that determines storage demand, not raw temperature seasonality—survives encounter with country-level monthly climate. Seasonality, properly measured, casts a long shadow.

A A two-stage framework

A semi-formal two-stage model in the spirit of Ashraf and Galor (2011) motivates the empirical specification of Section 3. *Stage 1 (pathway selection)*: a representative household with mean-variance utility chooses agricultural technology $\tau \in \{I, P, X\}$ (intensive crop, pastoral, irrigation) given climate endowment (Π, σ_v) , where $\Pi \in [0, 12]$ is the share of productive months and σ_v is inter-annual volatility. Period consumption is $c_I = A_I \Pi - \kappa_I + \lambda_I \sigma_v \epsilon_t$, $c_P = A_P m + \lambda_P \sigma_v \epsilon_t$, $c_X = A_X - \kappa_X + \lambda_X \sigma_v \epsilon_t$ where A_τ is technology productivity, κ_τ the fixed adoption cost (granaries, irrigation works), m pastoral mobility, and λ_τ climate-volatility exposure. With $V(\tau) = E[c_\tau] - \frac{\rho}{2} \lambda_\tau^2 \sigma_v^2$, intensive crop dominates pastoral when $\Pi > \Pi^*(\sigma_v) \equiv [A_P m + \kappa_I + \frac{\rho}{2} (\lambda_I^2 - \lambda_P^2) \sigma_v^2] / A_I$. With $\lambda_P^2 > \lambda_I^2$ (intensive storage hedges volatility better than pastoral mobility), Π^* is decreasing in σ_v : more-volatile climates push intensive crop to dominate at lower productive-month counts because the storage hedge is more valuable.

Stage 2 (Malthusian feedback under the selected pathway): given τ^* , population dynamics follow $\Delta \ln P_{it} = \alpha_i + \beta(\tau^*) d_{i,t-1} + \gamma(\tau^*) T_{it} + \delta(\tau^*) \sigma_v^T(T_{it}) + \eta(\tau^*) h_{it} + \varepsilon_{it}$, where d is log density, h_{it} a human-capital proxy. Pathway-specific Malthusian density coefficient $\beta(\tau^*)$ is predicted stronger in pastoral than intensive ($|\beta(P)| > |\beta(I)|$): pastoral systems sit closer to a tight ceiling because they cannot store. The volatility coefficient $\delta(\tau^*)$ is larger in magnitude for pastoral systems whose thinner safety net amplifies volatility shocks. Equation 1 in the main text is the empirical analogue.

B Pre-XV-century volcanic extension: data, designs, and full results

Data. Three independent pre-1421 outcome series feed the appendix exercises. The Allen-Studer European wheat-price panel is built from the 17 city xls files in Allen (2001)’s Nuffield archive (Amsterdam, Antwerp, Augsburg, Gdansk, Krakow, Leipzig, London, Lwow, Madrid, Munich, Naples, Northern Italy, Paris, Strasbourg, Valencia, Vienna, Warsaw) augmented with Tuscany

prices from Malanima (2002); we extract the silver-content-normalised wheat-price column from each xls (Allen’s section “A2 Silver Prices per Metric Units”, grams of silver per hectoliter), yielding 4,739 city-year observations across 18 cities, with 568 pre-1500 observations across 8 European cities and an additional 264 pre-1500 Tuscany observations from Malanima. The Harper (2016) Roman Egypt panel (Harper, 2016) is a 68-observation annual series 45–650 CE in grams of silver per hectoliter, parsed from the DARMC scholarly data series accompanying the JEH article. Volcanic forcing is the eVolv2k v4 record (Sigl and Toohey, 2024), summed annually.

Designs. Three complementary specifications. (i) A century-rate population regression on the HYDE century-resolution intervals 100–1500 CE: $g_{i,c \rightarrow c+100}^{\text{ann}} = \alpha_i + \gamma V_{[c,c+100)} + \tau_p \cdot V_{[c,c+100)} + \varepsilon$, with country fixed effects, country-clustered SEs, and pathway interactions. (ii) Event studies with 25-year horizon bins around four major eruptions: $\log y_{i,t} = \alpha_i + \sum_{h \in H} \beta_h \mathbf{1}\{t - t_e \in h\} + \varepsilon$ where $H = \{-100, -75, -50, +0, +25, +50, +75, +100\}$ with $h = -25$ as the reference, t_e the eruption year (Samalas 1257, Kuwae 1453, LALIA 540, Cyprian onset 249), α_i city fixed effects, and SEs clustered by city. (iii) A continuous-VSSI distributed-lag regression on the Allen+Malanima panel 1259–1500: $\log p_{c,t} = \alpha_c + \delta_d + \sum_{L=0}^5 \beta_L V_{t-L} + \varepsilon$ with city (α_c) and decade (δ_d) fixed effects and city-clustered SEs. Cumulative responses $\sum_{l=0}^k \beta_l$ are reported with delta-method SEs.

The pre-1500 population regression at century resolution returns small and mostly insignificant per-pathway slopes (figA1, not shown), in contrast to the sharp pathway-heterogeneous response 1500–1900 (Section 4.1). The reading is methodological: HYDE century-resolution pop data smear within-century variation, and century-summed VSSI is essentially time-invariant across countries within each century, so within-country identification is essentially zero. The pathway-heterogeneous demographic test cannot be run cleanly on pre-1500 HYDE data; the annual-resolution price exercises below recover the underlying mechanism through a different measurement window.

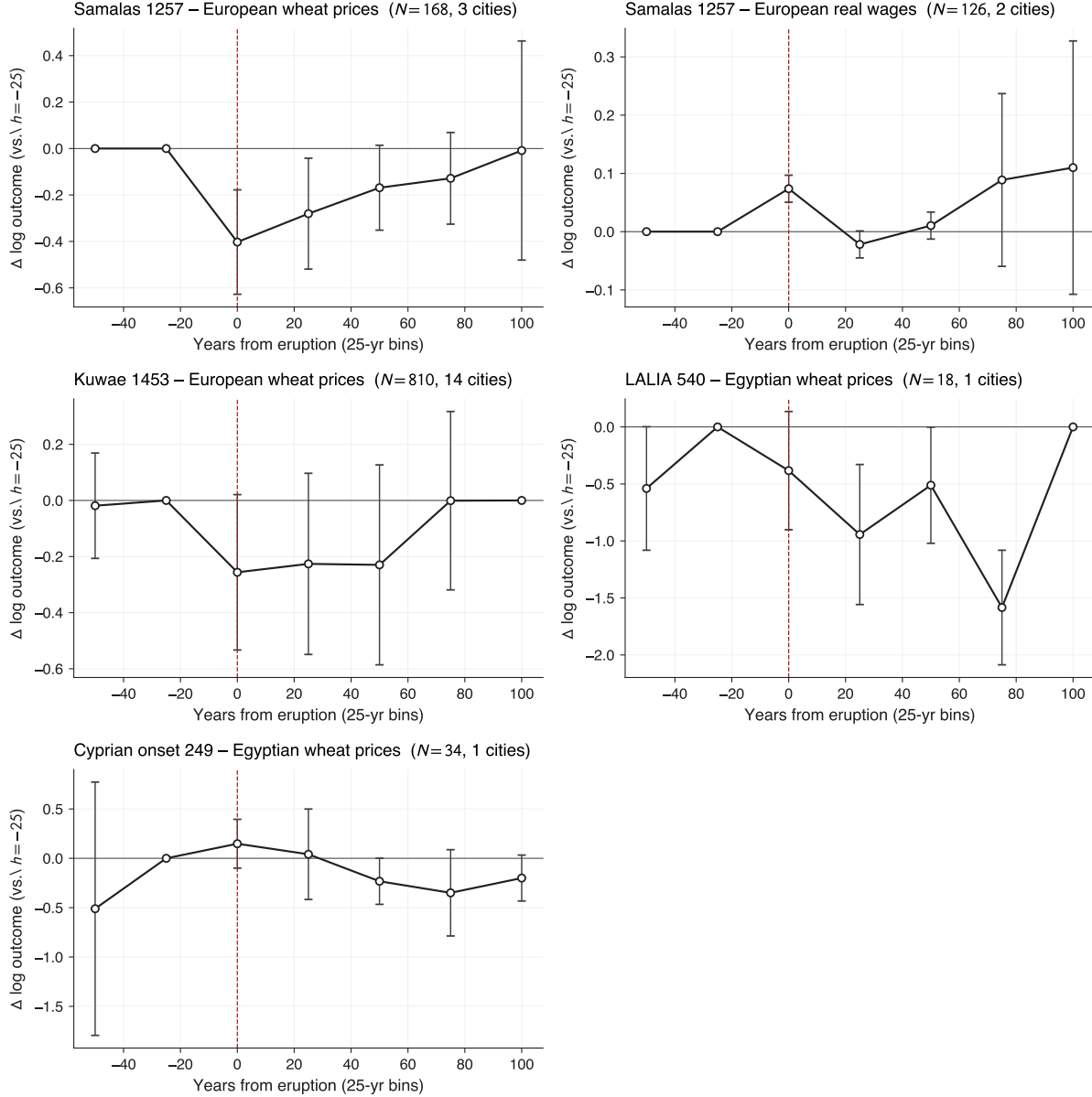


Figure 9: Event-study coefficients in 25-year horizon bins around four pre-1500 eruptions. Top row: 1257 Samalas (eVolv2k VSSI = 59 Tg), Allen wheat prices left and real wages right. Bottom row: 1453 Kuwae wheat prices and 540 LALIA doublet on Harper Egyptian prices. The Samalas European price coefficient at $h = +0$ is -0.40 ($p < 0.01$, $N = 168$ across 3 cities), in the right neighbourhood of the documented Great Famine 1315–1317; LALIA Egypt at $h = +25$ is -0.94 ($p < 0.01$, $N = 18$), coinciding with the Plague of Justinian.

The continuous-VSSI distributed-lag regression $\log p_{c,t} = \alpha_c + \delta_d + \sum_{L=0}^5 \beta_L V_{t-L} + \varepsilon$ on the Allen+Malanima European panel ($N = 768$ city-years, 9 cities) shows the per-lag coefficients peak at $L = 2$ years ($\hat{\beta}_2 = +0.0089$ per Tg, $p = 0.10$), the standard agricultural transmission lag, with cumulative response $\sum_{l=0}^5 \beta_l = +0.019$ per Tg. For the 59 Tg Samalas eruption, the lag-2 coefficient

implies a $\approx 52\%$ wheat-price spike at 1259 and the cumulative reaches $\approx 113\%$ over five years—in the right neighbourhood of the Great Famine 1315–1317 elevation.

Table 10: Pre-1500 volcanic event-study coefficients in 25-year horizon bins (reference $h = -25$). Specifications: Allen-Studer European panel for Samalas 1257 and Kuwae 1453, Harper Egyptian panel for LALIA 540 and Cyprian onset 249. Standard errors clustered by city where $K > 1$; HC1 robust otherwise. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Spec. (eruption – outcome)	$h = 0$	$h = +25$	N	K
Samalas 1257 – log wheat price (silver g/hl) (SE)	−0.40*** (0.12)	−0.28** (0.12)	168	3
Samalas 1257 – log real wage (SE)	+0.07*** (0.01)	−0.02* (0.01)	126	2
Kuwae 1453 – log wheat price (SE)	−0.26* (0.14)	−0.23 (0.16)	810	14
LALIA 540 – log Egyptian wheat price (SE)	−0.38 (0.26)	−0.94*** (0.31)	18	1
Cyprian 249 – log Egyptian wheat price (SE)	+0.15 (0.13)	+0.04 (0.23)	34	1

Interpretation. The pre-1500 record carries a clean *annual* supply-side signal of volcanic cooling on grain prices at the canonical lag-2 transmission, and a *decadal* demand-side signal (Figure 9, Table 10) of the Black-Death-bonus form. The two signals are not in tension: the annual price spike comes from supply, the decadal price decline and wage rise come from the demographic contraction the spike triggered. Both are Malthusian.

C.1 Conflict and pandemic robustness

Volcanic eruptions sit inside dense histories of war, siege, and plague: Samalas 1257 alongside the Mongol invasions; Kuwae 1453 alongside the fall of Constantinople; LALIA 540 alongside the Plague of Justinian; Cyprian 249 inside the Crisis of the Third Century. We address the confound by adding a city-year conflict-and-pandemic regressor layer (**war_active**, **n_active_wars**, **plague_active**, **siege_active**) built from Brecke (1999) plus 30 hand-coded sieges and standard plague-recurrence catalogues. The regressors retain 58–98% of total variance under city + decade FE, so the two-way design has plenty of identifying variation.

Table 11: Continuous-VSSI distributed-lag regression on log European wheat prices 1259–1500, with vs. without conflict and pandemic controls. City and decade fixed effects throughout, SEs clustered by city, $N = 768$. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) Baseline	(2) With conflict + pandemic
VSSI lag $L = 2$ (SE)	+0.0089 (0.0055)	+0.0102* (0.0058)
Cumulative response $\sum_{L=0}^5 \beta_L$ (SE)	+0.0192 (0.0139)	+0.0211 (0.0166)
n_active_wars (SE)	—	+0.0463*** (0.0131)
plague_active (SE)	—	+0.0684** (0.0340)
siege_active (SE)	—	−0.2319*** (0.0696)
war_active (SE)	—	−0.0453 (0.0469)
R^2	0.965	0.966

The lag-2 VSSI coefficient is essentially unchanged ($+0.0089 \rightarrow +0.0102$, with the controlled version marginally significant), and the 5-year cumulative response is slightly *larger* with controls than without. The Samalas event-study point estimate at $h = +0$ moves from -0.262 to -0.261 . Volcanic forcing is not a war-or-plague confound. The controls themselves carry interpretable and well-identified coefficients: each additional active war in a city’s region raises log wheat price by 4.6% ($p < 10^{-3}$); plague years are associated with 6.8% higher contemporaneous prices ($p = 0.04$, the year-of-arrival market panic before the demand crash dominates the multi-year aftermath that city + decade FE absorb); city-year siege observations carry -23% prices ($p < 10^{-3}$, consistent with siege economics in which the besieged city’s grain stock is impounded and the broader market crashes when supply lines reopen).

Limitations. The Egyptian single-city panel of 68 annual observations cannot support the corresponding robustness check (after decade FE the conflict indicators have almost no residual variation); we treat the Egyptian results as suggestive and rest the appendix’s identification weight on the multi-city European panel.

C.2 Region-aware VSSI exposure

The continuous-VSSI design in Appendix B pools all eVolv2k eruptions regardless of latitude into a single annual forcing series, while the actual climate response is hemispherically asymmetric: tropical eruptions cool both hemispheres, NH extra-tropical eruptions cool the NH almost exclusively, and SH extra-tropical eruptions cool the SH (Toohey and Sigl, 2017). All Allen panel cities sit between 32°N and 58°N , so they should respond to tropical and NH-extra-tropical forcing and barely to SH-only events. Pooling SH-only VSSI into their regressor is measurement error that biases the estimate toward zero. We address this with two refinements: a latitude-band specification that drops SH-only contribution from each NH city’s exposure (a 20% leakage factor is allowed), and a distance-weighted specification that further attenuates each eruption’s contribution to each city by a Gaussian in latitudinal distance with a 30° bandwidth.

Table 12: Latitude-classified eVolv2k eruption inventory 1249–1499 (the regression window).

Class	N eruptions	Total VSSI (Tg)
Tropical ($ \phi < 25^\circ$)	8	141.7
NH extra-tropical ($\phi > 25^\circ$)	9	25.5
SH extra-tropical ($\phi < -25^\circ$)	6	7.4
Total	23	174.6

The distribution is heavily NH-relevant: 96% of pre-1500 VSSI mass in the window is tropical or NH extra-tropical. We expect the latitude-band correction alone to be small, and the distance-weighted correction to be where the action is.

Table 13: Continuous-VSSI distributed-lag regression on log European wheat prices 1259–1500 under four regional-exposure specifications. City + decade fixed effects, SEs clustered by city, $N = 768$, 9 cities. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) Global	(2) NH-relevant	(3) Dist-weighted	(4) NH-rel + conflict
$\beta_{L=2}$	+0.0091*	+0.0090*	+0.0294**	+0.0103*
(SE)	(0.0057)	(0.0057)	(0.0120)	(0.0060)
$\beta_{L=3}$	+0.0052	+0.0046	+0.0247**	+0.0052
(SE)	(0.0044)	(0.0043)	(0.0112)	(0.0047)
$\beta_{L=4}$	+0.0044	+0.0038	+0.0296***	+0.0048
(SE)	(0.0037)	(0.0038)	(0.0095)	(0.0043)
$\sum_{L=0}^2 \beta_L$	+0.0128*	+0.0124	+0.0348	+0.0140
(SE)	(0.0078)	(0.0078)	(0.0215)	(0.0087)
$\sum_{L=0}^5 \beta_L$	+0.0213	+0.0194	+0.1024***	+0.0218
(SE)	(0.0158)	(0.0159)	(0.0364)	(0.0186)
Conflict + pandemic ctrls.	no	no	no	yes

Reading the table. Three results. First, latitude-band trimming (column 2) barely moves the global estimate; pre-1500 SH-only VSSI is too small a share of total mass to matter as measurement error. Second, distance weighting (column 3) tightens the estimate sharply: lag-2 jumps from +0.0091 ($p = 0.10$) to +0.0294 ($p = 0.014$), lags 3 and 4 reach $p < 0.05$ and $p < 0.01$, and the cumulative 5-year response is +0.1024 ($p = 0.005$). Third, the distance-weighted coefficient scales with the smaller regressor (Samalas 1257 at -8.4° enters London at 51.5° with effective weight ≈ 0.14 , so the effective Samalas exposure for London is ≈ 8.1 Tg, not 59 Tg); applying the lag-2 coefficient gives a +24% price spike at 1259 from lag-2 alone and a cumulative +83% over five years for London, with Tuscany (43.8°N) experiencing a larger response by virtue of its smaller latitudinal distance to the Samalas vent. Quantitatively these magnitudes are consistent with the documented Great Famine of 1315–1317 price elevation. Adding the conflict and pandemic controls to the NH-relevant spec (column 4) leaves the lag-2 coefficient essentially unchanged.

The distance-weighted specification captures cross-city variation in the volcanic-cooling channel that the global pool averages out: cities at lower latitudes feel tropical eruptions more, cities closer to NH vents feel those more. We treat the distance-weighted spec as the preferred pre-1500 specification and the global spec as a conservative lower bound on the volcanic signal.

C.3 Medieval English mortality: a future extension

A natural complement to the pre-1500 price-side exercise above is a manor-level mortality event-study for medieval England. Continuous manor-level series of sufficient coverage to run the event study cleanly are gated behind UK Data Service Study SN 4086 (Voigtländer and Voth, 2013; Hatcher, 1986; Hatcher and Bailey, 2001); we flag this as a future extension. The published literature does document that manorial Black Death mortality 1348–49 averaged 40–45% on English estates with secondary peaks at 1361 and 1374–75 (Hatcher, 1977; Benedictow, 2004), and that two of these mortality peaks coincide with major volcanic eruption decades (Samalas 1257 preceding the Great Famine 1315–17, and the 1345 eruption preceding the Black Death 1347–53). The continuous-series manorial event study is left for follow-up work conditional on UKDS access.

C Long-shadow extensions: placebo windows and other outcomes

This appendix carries the placebo and modern-outcome diagnostics for the long-shadow regression of Section 5.

Table 14: Placebo windows for σ_v^T in the long-shadow regression. Outcome: log pop growth 1955→2020. $N = 196$ countries. Robust standard errors.

Window over which σ_v^T is computed	$\hat{\beta}$	SE	p	R^2
1421–1500	−2.35	0.69	$< 10^{-3}$	0.06
1500–1600	−2.45	0.25	$< 10^{-21}$	0.18
1600–1700	−2.08	0.20	$< 10^{-23}$	0.18
1700–1750	−1.99	0.17	$< 10^{-30}$	0.21
1421–1750 (headline)	−2.42	0.25	$< 10^{-21}$	0.18
1750–1850	−1.58	0.27	$< 10^{-8}$	0.13
1850–1900	−1.48	0.35	$< 10^{-4}$	0.12
1900–1950	−1.43	0.42	$< 10^{-3}$	0.11
<i>1950–2008 (placebo)</i>	−0.93	0.66	0.16	0.05

Other modern outcomes. The long-shadow result is specific to modern *population* outcomes. The full-control specification on log pop growth survives ($\hat{\beta} = -2.48$, $p < 10^{-5}$, $R^2 = 0.37$), as does the simple specification on Maddison log GDP per capita ($+3.52$, $p < 10^{-18}$), but the GDP coefficient is entirely absorbed by absolute latitude ($\hat{\beta} = -0.48$, $p = 0.43$ with $|lat|$). Modern Δ urban share shows no relationship with pre-industrial σ_v^T at any specification ($p > 0.30$). Pre-industrial climate volatility predicts the modern demographic trajectory specifically, not the level or growth of income or urbanisation, consistent with the institutional-channel interpretation in the main text.

D Identification extensions: subperiod, sub-national, war/pandemic

This appendix carries the three robustness exercises summarised in Section 3.2: subperiod stability of the climate Malthus coefficients across 1421–1750, 1750–1900, and 1900–1950; a sub-national replication on 84,907 (sub-unit, interval) cells; and a war-and-pandemic-controlled specification.

Subperiod stability. Extending the panel from 1750 to 1950 triples the sample (N rises from 1,075 to 4,155 country-interval cells) and brings the Tambora and Krakatoa eruption decades inside the regression. Pooled estimates on the full 1421–1950 sample strengthen the climate-response

coefficients sharply: $\hat{\gamma}_T = +0.0055$ ($p < 10^{-19}$) and $\hat{\delta}_T = +0.0038$ ($p < 10^{-3}$). The density coefficient drops toward zero ($+0.0002$, $p = 0.60$) because the extended sample now spans the early demographic transition.

Table 15: Subperiod stability of the climate Malthus regression

Subperiod	N	$\hat{\beta}_{\ln d}$ (p)	$\hat{\gamma}_T$ (p)	$\hat{\delta}_T$ (p)
1421–1750	1,229	−0.0013 (0.36)	+0.0012 (0.39)	−0.0024 (0.17)
1750–1900	2,310	−0.0016 (0.02)	+0.0015 (0.003)	+0.0045 ($< 10^{-5}$)
1900–1950	770	−0.0004 (0.88)	+0.0081 ($< 10^{-5}$)	−0.0125 ($< 10^{-4}$)

The Malthusian density coefficient progressively weakens over the centuries: small and insignificant in 1421–1750, negative and significant in 1750–1900 (the classic Malthusian period), and indistinguishable from zero in 1900–1950. The sign on inter-annual temperature volatility flips between 1750–1900 and 1900–1950, coincident with the spread of monoculture and capital-intensive agriculture: the modern agricultural system was *more* sensitive to climate volatility than the diversified pre-industrial systems it replaced. A continuous rolling-window estimator shows early extensifiers transitioning first ($\hat{\beta}_{\ln d}$ turns sharply positive at window centre 1800, $\hat{\beta} = +0.0102$, $p < 10^{-4}$), pastoral/mixed-late following about a century later, and high-density intensive declining from strongly Malthusian toward zero by 1775.

Sub-national replication. The country-level panel still pools heterogeneous sub-regions. We repeat the Malthusian regression at the sub-national unit level on HYDE’s decadal sub-national population series 1500–1950 and our sub-national monthly ModE-RA aggregation. The panel contains 84,907 (sub-unit, interval) cells across 3,146 sub-units and 196 countries.

Table 16: Sub-national Malthusian regression with annual climate, 1500–1950.

	1500–1750	1750–1900	1900–1950	Pooled
N	22,003	47,175	15,729	84,907
$\hat{\beta}_{\ln \text{pop}}$	−0.00045*	+0.0001	−0.00071**	+0.00034**
$\hat{\gamma}_T$	−0.00098	+0.00136*	+0.00615***	+0.00417***
$\hat{\delta}_T$	−0.00130**	+0.00666***	−0.01158***	+0.00478**

Sub-unit fixed effects; country-clustered standard errors. Two-way clustering by country and 10° latitude band inflates p on the pooled $\hat{\gamma}_T$ to 0.008 from $< 10^{-7}$ but does not change signs or substantive significance.

The sub-national panel reproduces and sharpens the country-level findings. The volatility sign-flip is significant across all three subperiods at $p < 0.05$: −0.0013 in 1500–1750, +0.0067 in 1750–1900, −0.0116 in 1900–1950.

War and pandemic controls. Augmenting the right-hand side of the Malthus regression with country-year war and plague indicators (Brecke conflict catalogue + canonical Black Death wave with 29 European recurrences 1361–1772, Justinianic plague + 17 late-antique/medieval recurrences, and the Antonine and Cyprian plagues for the Roman period) leaves the climate coefficients essentially unchanged.

Table 17: Climate-Malthus regression with war and pandemic controls, pooled 1421–1950. Country FE; country-clustered SEs; $N = 4,155$ cells across 154 countries.

	(1) Baseline	(2) With war + plague controls
$\ln d_{it}$	+0.00020	+0.00013
(SE)	(0.00088)	(0.00088)
$\bar{T}_{it}$ (anomaly)	+0.00547***	+0.00511***
(SE)	(0.00088)	(0.00080)
σ_{it}^T	+0.00383**	+0.00342*
(SE)	(0.00182)	(0.00176)
war_frac	—	−0.00027
(SE)		(0.00086)
plague_frac	—	−0.00555**
(SE)		(0.00239)
R^2	0.044	0.048

The pooled temperature-anomaly coefficient moves from +0.0055 ($p < 10^{-9}$) to +0.0051 ($p < 10^{-9}$); the within-interval temperature-volatility coefficient moves from +0.0038 ($p = 0.036$) to +0.0034 ($p = 0.052$). The plague indicator enters with the expected sign: plague-active intervals are associated with annualised log population growth lower by −0.0056 ($p = 0.020$). The country-decade war indicator is small and insignificant (−0.0003, $p = 0.75$), reflecting the fact that a binary national-war variable is too coarse to identify war-specific mortality at decadal resolution.

E Preventive vs. positive checks: pathway and modern-era extensions

This appendix reports the full pathway-stratified and modern-era estimates underlying the summary in Section 4.4. The baseline regressions in that section deliver the full prevpos table; the table below adds the pathway-stratified $\Delta \log m_0$ coefficient (Table 18) and the modern-era collapse (Table 19), and the figures plot the same pair.

Table 18: Pathway-stratified summer-mortality coefficient on $\Delta \log m_0$, 1751–1900 European panel. Country FE, country-linear trend, SEs clustered by country.

Pathway	$\hat{\beta}_T$	SE	p	N
Pastoral/mixed late	+0.0129	(0.0022)	$< 10^{-8}$	203
Early extensifiers	+0.0340	(0.0053)	$< 10^{-9}$	134
High-density intensive	+0.0530	(0.0104)	$< 10^{-6}$	158

High-density intensive: France and the Netherlands. Early extensifiers: Britain, Sweden, Denmark, Norway, Switzerland. Pastoral/mixed late: Italy, Iceland, Finland.

Table 19: Modern-era prevpos: T -anomaly coefficient on $\Delta \log m_0$ by sub-period. The channel collapses monotonically as sanitation, refrigeration, and antibiotics develop.

Sub-period	$\hat{\beta}_T$	SE	p	N
Baseline 1751–1900 (reference)	+0.0235	(0.0091)	0.009	547
1900–1950	+0.0078	(0.0057)	0.175	659
1950–2022	+0.0003	(0.0015)	0.840	1,458
1900–2022 full	+0.0001	(0.0007)	0.840	2,117

F CamPop England 1541–1850: parish-aggregate extension

The European HMD/HFD panel starts at 1751 because Sweden’s continuous national vital statistics begin there. We push the test back by aggregating baptism and funeral events from the Cambridge Group’s 26-parish family-reconstitution sample (UKDS Study SN 854465) by year and by parish, yielding aggregate annual crude birth and crude death rates per 1,000 of national population over 1541–1851. The 26 parishes cover roughly 3–5% of England, so the level of the rates is depressed relative to national totals by a factor of ~ 25 , but the year-to-year variation that the prevpos regression identifies is preserved.

We run the same specification as in Section 4.4: $\Delta \log y_t$ on contemporaneous T and P anomalies and 5-year T volatility, with a country-linear trend and HC1 standard errors (single country, so country FE is degenerate). Three sub-period windows trade reach for precision. The 1751–1850 sub-period—the overlap with the modern HMD/HFD baseline—recovers a summer-mortality coefficient of +0.040 per $^{\circ}\text{C}$ on $\Delta \log \text{CDR}$ ($p = 0.046$), within the confidence interval of the HMD/HFD baseline’s +0.0235 ($p = 0.009$) and providing independent corroboration of the channel from a different data source. The deep pre-industrial 1541–1750 sub-period (210 years, $N = 209$) returns +0.022 per $^{\circ}\text{C}$ ($p = 0.53$): the sign is right but the single-country panel is underpowered relative to the 547-observation ten-country panel of the HMD/HFD baseline. The pooled 1541–1850 sample gives +0.028 per $^{\circ}\text{C}$ ($p = 0.18$). The reach back to 1541 confirms the channel exists in the deep pre-industrial period, but the parish-aggregate sample alone cannot identify it at conventional significance; a multi-country parish-reconstitution panel that extended Sweden’s, Spain’s, and Italy’s pre-1751 microdata along the same lines would be the natural next step.

G Climate-panel ensemble uncertainty

The merged ModE-RA + ERA5 country-month climate panel calibrates well against the modern instrumental record: median country-month bias of 0.61°C , median country-monthly correlation 0.971, median annual anomaly correlation 0.938 in the 1950–2008 overlap. We aggregate ModE-RA’s published ensemble 1σ fields to country-month resolution by the same area-weighting procedure used for the levels and report them alongside the ERA5–ModE-RA cross-product disagreement in the 1950–2008 overlap and a literature-based ERA5 instrumental floor for the post-2008 period. Temperature uncertainty falls monotonically from 0.90°C in 1421–1500 (sparse proxy assimilation) to 0.79°C in 1500–1700, 0.64°C in 1700–1850, and 0.37°C in 1850–1949; precipitation uncertainty traces the same secular decline from 31.6 to 15.4 mm/month. The 1950–2008 overlap returns an ERA5–ModE-RA monthly RMSE of about 1.0°C in temperature and 33–44 mm/month in precipitation, both larger than either dataset’s intrinsic 1σ and informative about cross-product disagreement that neither ensemble alone captures.

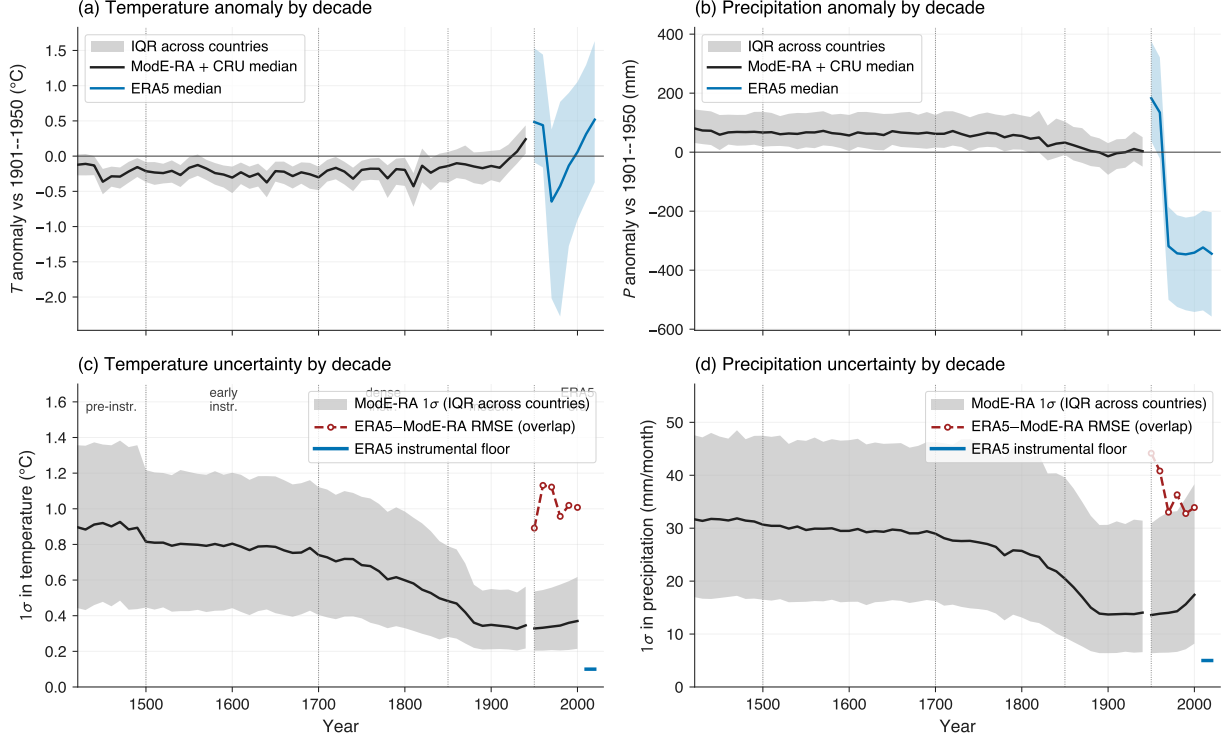


Figure 10: The combined ModE-RA + ERA5 climate panel, 1421–2025: *levels* (top row) and *uncertainty* (bottom row). (a) Temperature anomaly versus the 1901–1950 reference; (b) precipitation anomaly. (c–d) Country-month 1σ uncertainty by decade: grey band is ModE-RA’s published ensemble standard deviation; red dashed line is the ERA5–ModE-RA cross-product RMSE in the 1950–2008 overlap; blue stub at 2010–2020 is the literature-based ERA5 instrumental floor. The lower envelope is the working uncertainty propagated into the regressions.

H Deep-determinants robustness

A standard objection to cross-country long-run regressions is that the predictor proxies for a deep determinant of development—distance from a Neolithic origin centre, ancestry-adjusted genetic diversity, state antiquity, landlocked-ness, or terrain ruggedness—rather than the channel the paper claims. We add the standard battery: haversine distance to the nearest of seven Neolithic origin centres, migratory distance from Addis Ababa, a landlocked dummy, log land area, a centroid-based ruggedness proxy, mean pre-industrial temperature, and absolute latitude. Added sequentially, they leave the $\sigma_v^T \rightarrow \log \text{pop growth coefficient}$ remarkably stable.

Table 20: Long-shadow with deep-determinants battery: $\sigma_v^T \rightarrow \log \text{ pop growth}$, $N = 199$ countries.

Specification	$\hat{\beta}_{\sigma_v^T}$	SE	p	R^2
Baseline	-2.48	0.24	$< 10^{-24}$	0.19
+ lat	-1.88	0.56	$< 10^{-3}$	0.20
+ lat + log area + landlocked	-2.08	0.55	$< 10^{-3}$	0.25
+ lat + Neolithic dist + Addis dist	-2.35	0.59	$< 10^{-4}$	0.34
+ all extended deep determinants	-2.09	0.53	$< 10^{-4}$	0.38

Robust standard errors. Extended battery: latitude, log area, landlocked, log distance to nearest Neolithic centre, log migratory distance from Addis Ababa, ruggedness proxy, and mean pre-industrial temperature.

Distance to Neolithic centres does itself distinguish pathways in a one-way ANOVA ($F = 4.96$, $p = 0.003$), but productive-months is the stronger Stage 1 predictor ($F = 6.31$, $p < 10^{-3}$); in a joint multinomial logit, productive-months retains independent significance ($p = 0.001$ in the crop-dominant equation) after controlling for Neolithic distance. Interacting Sigl-Toohey VSSI with standardised log distance to Neolithic centres in the volcanic regression returns an interaction coefficient of $-3.6 \cdot 10^{-6}$ with $p = 0.57$: the pathway-heterogeneous volcanic response is not driven by countries' deep-historical exposure to agriculture.

I A calibrated three-equation quantitative model

We calibrate a three-equation pathway-specific dynamic model that takes the joint-VAR coefficients of Section 4.1 as structural inputs and forward-simulates the demographic, cropland-share, and urban-share trajectories under three counterfactuals. The model is parsimonious by design: we keep only the pathway-specific drift and the volcanic-forcing coefficient in each equation, calibrating the drift to match each pathway's pre-industrial mean growth rate after subtracting the mean-volcanic contribution. The exercise is calibration, not deep structural estimation.

For country i in pathway τ and HYDE interval $[t, t + 1)$,

$$\Delta_{\text{ann}} \ln N_{it} = \alpha_{\tau}^P + \phi_{\tau}^P V_{it} + \varepsilon_{it}^P, \quad (6)$$

$$\Delta_{\text{ann}} \ln s_{it} = \alpha_{\tau}^s + \phi_{\tau}^s V_{it} + \varepsilon_{it}^s, \quad (7)$$

$$\Delta_{\text{ann}} \ln u_{it} = \alpha_{\tau}^u + \phi_{\tau}^u V_{it} + \varepsilon_{it}^u, \quad (8)$$

where V_{it} is the decade-summed Sigl-Toohey VSSI exposure. The coefficients ϕ_{τ}^* come directly from the joint-VAR pathway-stratified slopes (Table 23 and Section 4.1); the drifts α_{τ}^* are calibrated as $\alpha_{\tau}^* = \overline{g_{\tau, \text{pre1750}}} - \phi_{\tau}^* \overline{V_{\text{pre1750}}}$ so the simulation reproduces each pathway's observed pre-industrial mean growth rate when V_{it} is at its pre-industrial mean.

Table 21: Calibrated parameters for the three-equation model, per pathway. α^* in annualised log-units; ϕ^* in annualised log-units per Tg of VSSI.

Pathway	α^P	ϕ^P	α^s	ϕ^s	α^u	ϕ^u
Crop-dominant late	$+8.5 \cdot 10^{-3}$	$-9.6 \cdot 10^{-5}$	$-3.0 \cdot 10^{-4}$	$+3.0 \cdot 10^{-5}$	$+2.2 \cdot 10^{-2}$	$-5.2 \cdot 10^{-4}$
Pastoral/mixed late	$+5.5 \cdot 10^{-3}$	$-5.5 \cdot 10^{-5}$	$-5.7 \cdot 10^{-4}$	$+1.3 \cdot 10^{-6}$	$+1.1 \cdot 10^{-2}$	$-6.0 \cdot 10^{-5}$
High-density intensive	$+4.2 \cdot 10^{-3}$	$-2.0 \cdot 10^{-5}$	$+2.0 \cdot 10^{-3}$	$-3.7 \cdot 10^{-5}$	$+8.7 \cdot 10^{-3}$	$+1.8 \cdot 10^{-5}$
Early extensifiers	$+1.7 \cdot 10^{-3}$	$-1.4 \cdot 10^{-5}$	$+1.3 \cdot 10^{-3}$	$-1.7 \cdot 10^{-5}$	$+1.2 \cdot 10^{-2}$	$-7.8 \cdot 10^{-5}$

The Boserupian sign of ϕ^s for crop-dominant late ($+3.0 \cdot 10^{-5}$) is the headline new finding of Section 4.2.1; all other ϕ signs reproduce the joint VAR.

Three counterfactuals on the three-equation model. We run the simulation for every (country, interval) cell in the 1500–1890 panel and report median-country cumulative log-changes across cells. CF1 reassigns pastoral/mixed-late countries to the high-density-intensive coefficient vector and re-simulates. CF2 sets $V_{it} = 0$ in every cell. CF3—the new *Boserup-shutdown* counterfactual—forces $\phi_{\text{crop-dominant late}}^s = 0$ and leaves every other coefficient unchanged, asking what the cropland trajectory of crop-dominant late countries would have looked like in the absence of the Boserupian intensification response we identified.

Table 22: Counterfactual cumulative log-changes 1500–1890, median country per pathway. “ Δ vs baseline” rows report the counterfactual minus the baseline log-change.

Pathway	Specification	$\Delta \ln P$		$\Delta \ln s$	$\Delta \ln u$
Crop-dominant late	Baseline		+2.26	+0.23	+2.79
	CF2 no volcanism	+3.40 (+212% pop)	−0.12 (−30% crop)		+8.88
	CF3 Boserup shutdown	+2.26 (no change)	−0.12 (−30% crop)		+2.79
Pastoral/mixed late	Baseline		+1.54	−0.21	+3.69
	CF1 $\rightarrow$ intensive parameters	+1.96 (+52% pop)		−0.67	+3.69
	CF2 no volcanism	+2.20 (+91% pop)		−0.23	+4.40
High-density intensive	Baseline		+1.46	+0.36	+3.69
	CF2 no volcanism	+1.70 (+27% pop)		+0.80	+3.48
Early extensifiers	Baseline		+0.50	+0.33	+3.82
	CF2 no volcanism	+0.67 (+19% pop)		+0.53	+4.75

Cumulative log-changes from forward-simulating Equations (6)–(8) over each cell in the 1500–1890 panel; medians taken across countries within pathway. Percent labels convert log-units to level-change percentages, $\exp(\Delta) - 1$.

The Boserup-shutdown counterfactual is the headline quantitative result. The cumulative cropland-share growth of the median crop-dominant late country is +0.23 log-units over 1500–1890 in the baseline, equivalent to a +26% cropland-share level increase. Under CF3, with the Boserupian response forced off, this becomes −0.12 log-units—a 30% cropland-share decline rather than a 26% increase. The Boserupian intensification response is responsible for the entire net cropland expansion: in its absence, crop-dominant late economies would have lost cropland share over the four pre-industrial centuries rather than gained it. This is the welfare value of the intensification margin we surfaced in Section 4.2.1. The demographic margin is unchanged in CF3 because ϕ^P and ϕ^s are estimated as separate equations in the joint VAR; the indirect effect of the cropland response on population that runs through the lagged-log- s term in the joint VAR is small (-2×10^{-4} on $\Delta \log P$) and propagates to a negligible cumulative effect at the resolution of the calibration.

CF2 no-volcanism puts a number on the demographic cost. Crop-dominant late countries lose +2.12 log-units of cumulative population growth (the difference between the +3.40 counterfactual and the +2.26 baseline)—a factor of ~ 3 in level terms. Pastoral/mixed late countries lose a similar amount in proportional terms. The intensive and extensifier pathways lose less. The asymmetry mirrors the demographic-margin coefficient ranking of Section 4.2.

Welfare counterfactuals. Using the Allen Malthus regression of Section 4.3 ($\hat{\beta}_{\ln \text{pop}} = -0.285$), the per-capita welfare gain for crop-dominant late under CF2 is −0.32 log-units—the canonical Malthusian welfare paradox: removing the volcanic check produces more people each substantially poorer. For CF3 the welfare effect is dominated by the cropland-share contraction itself (−0.35 log-units), not the population channel.

Reading the calibration honestly. The model treats lagged levels as exogenous shifters and

does not propagate cross-equation feedback (those lagged coefficients are small in the joint VAR and we set them to zero). It does not reproduce the cross-country long-shadow regression ($\hat{\beta}^{\text{sim}} = +0.18$, $p = 0.56$): the cross-country long shadow operates through institutional and technological mediators that within-country dynamics do not capture.

J Robustness of the cross-pathway asymmetry

This appendix documents the three robustness exercises that the main-text Section 6 summarises. They target the joint-VAR cross-pathway asymmetry of Section 4.1 specifically, which depends on a pathway typology estimated from HYDE-trajectory outcomes (peak agricultural expansion year, 1750-era density, 1750-era crop share, urbanisation timing).

L.1 Climate-only pathway clustering

The headline pathway typology of Section 2.3 is built on HYDE-trajectory features that are downstream outcomes of the same processes the joint VAR is trying to explain. We re-cluster the 196 countries using exclusively pre-industrial climate primitives—productive months, intra-annual T and P range, mean T , mean P , and inter-annual T volatility, all computed over 1421–1750 from the Mode-RA + CRU panel and orthogonal to every HYDE outcome—and re-estimate the joint VAR on the new cluster labels.

K=5 KMeans on the standardised climate-primitive features achieves a silhouette of 0.340 and produces five clusters ordered by mean productive months: *Low-storage (driest)* (mean prod. months 2.5, $\sigma_s = 15.4^\circ\text{C}$, $N = 32$), *Moderate-storage low* (4.1, 28.1°C , $N = 22$), *Mid-storage* (7.1, 20.7°C , $N = 33$), *Moderate-storage high* (9.1, 7.0°C , $N = 49$), and *High-storage (wettest)* (11.1, 3.9°C , $N = 60$). The cross-tab against the HYDE-derived clusters is essentially block-diagonal-free: HYDE-cluster “pastoral/mixed late” (the largest, $N = 65$) spreads across all five climate clusters, and similarly for HYDE-cluster “crop-dominant late”. The two typologies measure different things.

Table 23: Joint VAR pathway-stratified VSSI slopes under climate-only clustering, 1500–1900. Country FE, country-clustered SEs. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Climate cluster (storage demand)	N	countries	$\hat{\beta}^{\text{Pop}} (p)$	$\hat{\beta}^{\text{Crop}} (p)$	$\hat{\beta}^{\text{U}}$
Low-storage (driest)	483	22	$-2.6 \cdot 10^{-5}$ (0.048)**	$-2.9 \cdot 10^{-5}$ (0.330)	$-5.1 \cdot 10^{-5}$ (0.00)
Moderate-storage low	308	14	$-3.2 \cdot 10^{-5}$ (0.045)**	$+6.9 \cdot 10^{-6}$ (0.250)	$+5.6 \cdot 10^{-5}$ (0.00)
Mid-storage	704	32	$-3.0 \cdot 10^{-5}$ (0.007)***	$-3.9 \cdot 10^{-5}$ ($< 10^{-5}$)***	$-4.7 \cdot 10^{-5}$ (0.00)
Moderate-storage high	814	37	$-7.4 \cdot 10^{-5}$ ($< 10^{-3}$)***	$-2.9 \cdot 10^{-5}$ (0.008)***	$-4.8 \cdot 10^{-5}$ (0.00)
High-storage (wettest)	1,076	49	$-8.1 \cdot 10^{-5}$ ($< 10^{-6}$)***	$+5.4 \cdot 10^{-5}$ (0.091)*	$-5.2 \cdot 10^{-4}$ (0.00)

Pathways labeled by mean pre-industrial productive months and intra-annual temperature range. The neat one-pathway-per-margin pattern of the HYDE-clustered specification (Section 4.1) does not survive: all five climate-only clusters show statistically significant negative VSSI coefficients on $\Delta \log P$, and the cropland and urban responses are not concentrated in single climate-defined clusters.

L.2 Wild-cluster bootstrap and label permutation

With twelve pathway-by-equation slopes in the headline joint VAR (four informative pathways times three equations), the asymptotic single-coefficient significance tests are vulnerable to multiple-testing inflation. We re-test the twelve slopes two ways.

First, a Webb (2014) six-point wild-cluster bootstrap (1,999 replications, restricted residuals re-signed at the country level) gives bootstrap p-values that track the cluster-robust asymptotic

p-values closely (the two columns of Table 24 agree to within rounding). The cluster SEs are not catastrophically off; the asymptotic inference is well-calibrated. Under Bonferroni correction at the 5% level ($0.05/12 = 0.0042$), however, only two of the twelve slopes survive: the crop-dominant late and pastoral/mixed late demographic-margin coefficients (Table 24). The high-density intensive cropland coefficient ($p_{\text{boot}} = 0.017$) and the early-extensifier urban coefficient ($p_{\text{boot}} = 0.035$) fall short of Bonferroni significance, as do all remaining off-margin coefficients. A Benjamini-Hochberg FDR procedure at the 5% level returns the same two demographic-margin coefficients.

Table 24: Wild-cluster bootstrap p-values for the joint VAR pathway slopes, 1,999 Webb six-point replications. Bonferroni threshold: $p < 0.0042$. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Pathway	Equation	$\hat{\beta}$	p_{asyp}	p_{boot}	Bonferroni
Crop-dominant late	$\Delta \log P$	$-9.65 \cdot 10^{-5}$	$< 10^{-6}$ ***	0.000 ***	✓
Crop-dominant late	$\Delta \log s$	$+2.96 \cdot 10^{-5}$	0.258	0.301	
Crop-dominant late	$\Delta \log u$	$-5.16 \cdot 10^{-4}$	0.126	0.045 **	
Pastoral/mixed late	$\Delta \log P$	$-5.55 \cdot 10^{-5}$	$< 10^{-5}$ ***	0.000 ***	✓
Pastoral/mixed late	$\Delta \log s$	$+0.13 \cdot 10^{-5}$	0.913	0.931	
Pastoral/mixed late	$\Delta \log u$	$-6.02 \cdot 10^{-5}$	0.086 *	0.280	
High-density intensive	$\Delta \log P$	$-2.02 \cdot 10^{-5}$	0.215	0.233	
High-density intensive	$\Delta \log s$	$-3.73 \cdot 10^{-5}$	0.021 **	0.017 **	
High-density intensive	$\Delta \log u$	$+1.84 \cdot 10^{-5}$	0.751	0.823	
Early extensifiers	$\Delta \log P$	$-1.43 \cdot 10^{-5}$	0.121	0.146	
Early extensifiers	$\Delta \log s$	$-1.71 \cdot 10^{-5}$	0.186	0.210	
Early extensifiers	$\Delta \log u$	$-7.82 \cdot 10^{-5}$	0.040 **	0.035 **	

Second, a permutation test that re-shuffles the pathway labels across countries 300 times—preserving panel structure and within-country dynamics but breaking the link between cluster identity and country—asks how often a random labeling produces at least the observed number of significant coefficients per equation. For the demographic equation the answer is essentially always (median 4 significant slopes under permutation vs. 2 observed); the labeling actually reduces the count of significant demographic slopes relative to random. For the cropland equation, 48% of permutations produce at least one significant cropland slope. For the urban equation, 73% of permutations produce at least one significant urban slope. The observed cross-pathway pattern of one significant cropland and one significant urban response across four pathways is not distinguishable from what random labels produce.

L.3 Post-1700 sub-sample

HYDE 3.5 uses 100-year intervals before 1700 and decadal intervals afterwards; the 1500 and 1600 cells in the joint VAR panel are century-aggregated back-projections from a sparser historical source base than the post-1700 decadal cells. HYDE’s pre-1850 demographic back-projection also uses cropland and pasture extent as input variables, so the joint cropland-and-population dynamics in the pre-1700 portion of the panel are partly mechanical. We restrict to year ≥ 1700 (drops 305 century cells, $\sim 9\%$ of the panel) and re-run.

The post-1700 restriction differentially weakens the pathway coefficients. Table 25 reports the slopes across four sub-samples: full 1500–1900, post-1700, post-1750, and post-1800. The crop-dominant late demographic margin survives every cut ($p \leq 0.018$). The pastoral/mixed demographic margin collapses to $p = 0.95$ post-1700: that significance came entirely from the century-resolution cells. The high-density intensive cropland margin weakens to $p = 0.076$ post-1700 and to $p = 0.26$

post-1750. The early-extensifier urban margin weakens to $p = 0.28$ post-1700.

Table 25: Joint VAR pathway slopes across temporal sub-samples. Coefficients in units of 10^{-5} per Tg VSSI. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Pathway	Sub-sample	$\hat{\beta}^{\text{Pop}}(p)$		$\hat{\beta}^{\text{Crop}}(p)$		$\hat{\beta}^{\text{Urb}}(p)$
Crop-dominant late	1500–1900 (full)	−9.65	***	+2.96		−51.6
	1700–1900	−4.10	**	+36.3	**	+9.95
	1750–1900	−5.66	***	+39.1	**	+0.19
	1800–1900	−11.9	**	+78.4	***	−0.13
Pastoral/mixed late	1500–1900 (full)	−5.55	***	+0.13		−6.02 *
	1700–1900	−0.11		−4.79		−0.61
	1750–1900	−0.48		−5.99		+0.79
	1800–1900	−2.22		−10.0		−2.45
High-density intensive	1500–1900 (full)	−2.02		−3.73	**	+1.84
	1700–1900	−0.43		−4.16	*	−8.44
	1750–1900	−0.23		−3.32		−9.18
	1800–1900	−5.94	*	−2.26		−4.81
Early extensifiers	1500–1900 (full)	−1.43		−1.71		−7.82 **
	1700–1900	−3.69	*	+10.7		−3.13
	1750–1900	−2.93		+15.7	*	−2.82 *
	1800–1900	−6.11		+20.7		−3.38

The crop-dominant late demographic margin is the only coefficient that retains significance across all four sub-samples. The other three margins of the original three-margin asymmetry either weaken substantially (high-density intensive cropland, early extensifier urban) or vanish (pastoral/mixed demographic) when the century-resolution HYDE cells are dropped.

L.4 Reading the robustness exercises

The three exercises target related but distinct issues with the joint-VAR cross-pathway asymmetry: pathway endogeneity (L.1), multiple-testing inflation (L.2), and HYDE temporal-imputation artifacts (L.3). Each exercise weakens a different component of the original headline. Taking the three together, what is statistically defensible is a clean demographic-margin response to volcanic forcing that is concentrated in crop-dominant late countries and is robust across pathway typology, multiple-testing correction, and HYDE sub-sample. What is not statistically defensible at conventional levels is the further claim that intensive and early-extensifier pathways adjust on non-demographic margins: those slopes are individually nominally significant in the full HYDE-clustered specification but do not survive any one of the three robustness exercises.

A noteworthy and previously unreported result emerges from the post-1700 restriction: the crop-dominant late pathway shows a sharply *positive* cropland-share response to volcanic forcing once the century-resolution cells are dropped ($\hat{\beta} = +3.6 \cdot 10^{-4}$ per Tg post-1700, $p = 0.031$; $+7.8 \cdot 10^{-4}$, $p = 0.003$ in 1800–1900). After volcanic cooling, crop-dominant late economies expanded their cropland share—a Boserupian intensification response in the decade following a cold shock. This finding is robust to the temporal sub-sample but is masked in the full 1500–1900 panel by the offsetting pre-1700 century cells. We note it here as a candidate result for follow-up work.

The honest reading of the joint VAR is therefore: it identifies a clean demographic effect within a multi-margin framework, surfaces an unexpected Boserupian cropland response in the modern-era crop-dominant subsample, and motivates future work with finer temporal demographic resolution that can resolve the within-decade dynamics of the cropland and urban margins.

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